SiMPL Electronic Batch Record — PI Sheet | **New vs Old**

AZD0543 — Virus Inactivation + Concentration & Diafiltration (2000 L, Process 1)

Material / Product	AZD0543 — Antibody-Drug Conjugate (DSP, Process 1, GPF, 2000 L)
Records covered	PN 8012441 Virus Inactivation (Low pH) + PN 8012475 Concentration & Diafiltration (Large Skid UF/DF)
MABR / MPR	MABR-0027570 (VI) | PN 8012475 (C&D)
Process Order No.	____________
Batch #	____________
Plant / Facility	GPF
Effective Date	____________

How to read this: the digital EBR is assembled from XStep building blocks. Each card shows the original MPR section it replaces ("Original BR: §___") and is tagged NEW (built new), VARIANT (a new variant of an existing SMPL XStep), or REUSE (reuses an existing DE1 100 SMPL XStep as-is — shown as a summary, no new mock-up). **Beneath each XStep card is the verbatim full text of the old paper section(s) it must fulfill** — every instruction with the exact fields the operator recorded — so you can confirm, line by line, that the XStep covers the requirement.

8 new + **0** variant + **29** reuse = **37** XSteps for both records.

XStep ↔ Original MPR Crosswalk

EBR Phase	XStep	Type	Original MPR Section
Process Order Information & Front Matter	SMPL: Order Header Details	R	Header
	SMPL: Signature Table	R	Personnel ID
	SMPL: Display BOM	R	§4
	SMPL: Room & Equipment Assign	R	§7 (VI 7.3-7.4 / C&D 7)
	SMPL: Additional Manufacturing Supplies	N	§4
	SMPL: Additional Solution Batches (Solution Switch)	N	§5
	SMPL: Process Notes & Global Limits (Acknowledgement)	R	§6
Reusable Building Blocks (instantiated across BOTH records)	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	N	VI 7.5 · C&D 7.4-7.6
	SMPL: Product Vessel Weigh	R	VI §7/8/15 · C&D §9/11/16/17/18
	SMPL: Calc — Three Columns (Value1 op Value2 = Result)	R	VI 7.12 / 8.10 / 14.15 · C&D §7/16/17
	SMPL: Multi-Variable Calculation (3-input / ranged)	R	C&D §12/16/17
	SMPL: Product Mixing	R	VI §15 · C&D §9/16/17
	SMPL: Product Sampling & DLIMS Submission	R	VI §15/Att1 · C&D §8/14/16/17/Att3
	SMPL: Product pH / Conductivity / Temperature	R	VI 7.1/15.5 · C&D 8.12/14.11/17
	SMPL: Hold Time Table	N	VI §19 (Att3) · C&D §9/§25 (Att5)
	SMPL: Product Recirculation Worksheet	R	VI Att6 · C&D Att7
	SMPL: Transfer Tubing / Hosing Information	R	VI Att5 · C&D Att1
	SMPL: Product Storage & Cross-Record	N	VI 15.8 · C&D §9/§18
	SMPL: Instruction & Sign-off	R	VI §8/16/20 · C&D §11/20
	SMPL: Yield Calculations	R	VI Att4 · C&D Att8
	SMPL: Comments / Deviations Section	R	VI Att7 · C&D Att9
	SMPL: Non-Routine Sampling Record	R	VI Att2 · C&D Att4

Virus Inactivation — PN 8012441 (Low pH, 2000 L)	SMPL: VI Treatment Vessel Setup	N	§8
	SMPL: Iterative pH Titration Table (Acid / Base)	N	§10 / 12 / 14
	SMPL: Incubation Timing & Incubated Sample (block stack)	R	§11 / 13
	SMPL: Starting / Transfer Temperature Check (block stack)	R	§7.14 / 9
	SMPL: Store VI Product & Hold-Time Link (block stack)	R	§15.8
Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	R	§7.2 / §11 / Att6
	SMPL: Run Skid Recipe (block stack)	R	§8/10/13/15/19
	SMPL: Minimum Membrane Surface Area (block stack)	R	§7.7
	SMPL: Process Input Calculations (calc-block stack)	R	§12
	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	N	Att2 (§13-15)
	SMPL: Conditional Continued Diafiltration (block stack)	R	§14
	SMPL: Recovery Calculations & Execution (block stack)	R	§16
	SMPL: Dilution Decision & Calculations (block stack)	R	§17
	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	R	§18
	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	R	§19 / §20

Process Order Information & Front Matter

SMPL: Order Header Details REUSE

Original BR: Header

Product / PN, process order numbers, recorded by — standard SAP order header. Reused as-is.

Reuses: **DE1 100: Display Process Order Header Data (AZ Components) / SXS: SIMPL Order Header**

SMPL: Signature Table REUSE

Original BR: Personnel ID

Personnel identification: print name, signature, initials for everyone executing the record. Reused as-is.

Reuses: **DE1 100: SMPL Header: Signature Table**

SMPL: Display BOM REUSE

Original BR: §4

Bill of Materials — equipment (VI Treatment Vessel / UF-DF skid, meters, scale, mixer, FIT) and components / filters / bags with SAP consumption. Reused as-is.

Reuses: **DE1 100: Display BOM Material Table**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 15 GROUP(S)):

§4 Bill of Materials – Equipment and Components VIRUS INACTIVATION

- (records)

Part No./ Equipment ID | Quantity | Chromatography Skid | Mixer Drive | Peristaltic Pump | Quattroflow Pump | pH / Conductivity meter | Thermometer | Floor Scale / Benchtop Balance | Flow Meter | Dip tube | Clean Hosing | 200 L Portable Tank | 200 L Portable Tank | Fixed Tanks | 1500 L Product Hold Tank | 2000 L Product Hold Tank | 2000 L Product Hold Tank | 2500 L Product Hold Tank | 4000 L Product Hold Tank | Product Filters for Portable/Fixed Product Tanks | Filter – 10" SHC Cartridge 0.2 µm Hydrophilic | Filter – 5" SHC Cartridge, 0.2 µm Hydrophilic | Filter: Fluorodyne EX Grade EDF Cartridges (10 in.) | Vent Filters for Portable/Fixed Product Tanks | Filter – Aervent 0.2 µm, Hydrophobic, PTFE Membrane (5 in) | Filter – Aervent 0.2 µm, Hydrophobic, PTFE Membrane (10 in)

- 4 Bill of Materials – Equipment and Components (continued)

Part No./ Equipment ID | Quantity | Product Filters for Bags | Filter – SHC 0.5+0.2 µm Capsules with C-Flex (3") | Filter – SHC 0.5+0.2 µm Capsules with C-Flex (10") | Filter – 30 in., Opticap® XLT 30 Capsule Filter with Millipore Express® SHC Hydrophilic Membrane, 0.5/0.2 µm | Filter – 10 in., Opticap® XLT 10 Capsule filter with Millipore Express® SHC membrane, 0.5/0.2 µm | Mixing Bags for Product Filtration Only | LevTech Mix Bag (50 L) | LevTech Mix Bag (100 L) | LevTech Mix Bag (200 L) | LevTech Mix Bag (300 L) | LevTech Mix Bag (560 L) | Mixing Bags for Product Filtration and Viral Inactivation Treatment | Custom MagMix Bag (200 L) | Custom MagMix Bag (300 L) | Custom MagMix Bag (370 L) | Custom MagMix Bag (560 L) | Bags for WFI | Bags - Stedim 3D Bags with C-Flex (500 L) | Bags - Stedim 3D Bags with C-Flex (300 L) | Bags - Stedim 3D Bags with C-Flex (200 L)

- * = Materials with bolded filter part numbers require autoclaving prior to use.
- Additional Manufacturing Supplies Used in Process
- If additional materials (filter, bag, etc.) need to be used during processing:
- Prior to use, ensure the material is acceptable for use based on Section 4.
- Record the applicable material information.
- Document the reason for the change in Attachment 7: Comments Section.

§4 Bill of Materials – Equipment and Components CONC / DIAFILTRATION

- (records)

Part No./ Equipment ID | Quantity | PALL Large Scale UF/DF Skid | Mixer Drive | Peristaltic Pump | Stir Plate | Conductivity Meter | pH meter | Thermometer | Floor Scale/ Benchtop Balance | Filter Integrity Tester | Stainless Steel Dip tubes | Sight glasses | 5 L Glass Bottle | 10 L Glass Bottle | 20 L Glass Bottle | Part No./ Equipment ID | Quantity | UFDF Membranes | PELLICON 3 30kDa (C Screen) 0.11 m2 | PELLICON 3 30kDa (C Screen) 0.57 m2 | PELLICON 3 30kDa (C Screen) 1.14 m2 Pellicon Manifold Support Plate | Portable Formulation Tanks | 100 L Diafiltration / Concentration Tank | 200 L Diafiltration / Concentration Tank | 100 L Diafiltration / Concentration Tank | 100 L Diafiltration / Concentration Tank | Vent Filters for the Diafiltration / Concentration Tanks | Filter – Aervent, 0.2 µm, Hydrophobic, PTFE Membrane (5 in.) | Filter – Aervent, 0.2 µm, Hydrophobic PTFE Membrane (10 in.) | Product Filters for Filtration | Filter – 10-inch Opticap XLT Capsule w/ SHC Membrane, Sterile | Filter- 10 in., Opticap XLT 10 Capsule Filter with Millipore Express SHC Hydrophilic Membrane, 0.5/0.2 m | Filter – SHC 10" 0.5+0.2 um Capsule with C-flex | Filter – SHC 3" 0.5+0.22 um Capsule with C-flex | Filter – 30 in., Opticap XLT 30 Capsule Filter with Millipore Express SHC Hydrophilic Membrane, 0.5/0.2 m | Filter- 4 in., Opticap XL Capsule Filter with Hydrophilic Durapore Membrane, 0.22 m | Filter- Millipak 200 0.22 µm, Hydrophilic Gamma Irradiated | Bags for Product Mixing/Collection | LevTech Mix Bag (50 L) | LevTech Mix Bag (100 L) | LevTech Mix Bag (200 L) | LevTech Mix Bag (300 L) | LevTech Mix Bag (560 L) | Bags for WFI | Bags – Stedim 3D Bags with C-flex (300 L) | Bags – Stedim 3D Bags with C-flex (200 L) | Stedim Bag- 50 L Flexel 2D Bag with C-flex

- * = Materials with bolded filter part numbers require autoclaving prior to use
- Additional Manufacturing Supplies Used in Process:
- If additional materials (filter, bag, etc.) need to be used during processing:
- Prior to use, ensure the material is acceptable for use based on Section 4.
- Record the applicable material information.
- Document the reason for the change in Attachment 9: Comments Section.

Serial No., if applicable

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Room & Equipment Assign REUSE

Original BR: §7 (VI 7.3-7.4 / C&D 7)

Record the process room and assign every instrument up front (pH/cond meter, thermometer, scale, mixer, stir plate, pump) with calibration check. Downstream steps retrieve via Get [Type] (SMPL: Equipment Select, which already carries Calibration Due Date). Reused as-is.

Reuses: **DE1 100: SMPL: Room/Equipment Assign + SMPL: Equipment Select (field-verified: has ID + Description + Calibration Due Date)**

▼ OLD PAPER BATCH RECORD — VERBATIM TEXT THIS REUSABLE XSTEP MUST FULFILL, HARVESTED FROM EVERY SECTION THAT RECORDS IT (8 SECTION(S), 12 GROUP(S)):

§7 Affinity Product Information / Pre-Processing Activities VIRUS INACTIVATION

- Record the room where this process will occur.

  Room Number

- Confirm that the pH meter has been standardized between pH 2.0 and pH 7.0 per SOP-0080297. Confirm that the conductivity meter has been standardized within the acceptable range per SOP-0080297.

  Meter ID

- Confirm that the thermometer is within its calibration interval. Record the equipment ID of the thermometer used.

  Thermometer ID

§14 Setup and Base Addition VIRUS INACTIVATION

- For Step Step 14.11 - Neutralization Treatment: After a target incubation time of 75 minutes, within a range of 60 – 240 minutes, begin transferring 1M Tris Base (A0101-D) into the VI Treatment Vessel: Do not stop agitation during treatment. For the base additions, follow SOP-0081015 if the VI Treatment Vessel is a MagMix bag. If the vessel is a tank, follow SOP-0107095. After each addition of A0101-D, mix the product for at least ten minutes prior to taking a sample for pH measurement. When pH rises above 4.0, recalibrate the pH meter to pH 4.0 and pH 10.0 and record the calibration information in Step 14.12. Continue adding A0101-D until the product pool is within pH range 7.2 – 7.6. Record the Final pH, Final conductivity, Temperature and End Time For Incubation in Step 14.11.

  Neutralization Information | End of Incubation Information | (A) pH at the end time of Incubation (3.4 – 3.6) | (B) Temperature at the end of Incubation (18 – 25 oC) | (C) End time for Incubation (Start Time of base addition) | Start Time of Incubation (Step 10.9 or 12.4) a | (D) Total Duration of Incubation (60-240 minutes) | minutes | Neutralization Treatment | Scale/Balance ID | Calibration Due Date | Gross weight of Treatment Vessel before addition (kg) | End time of base addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) b | Total net weight of buffer added (kg) c | Time of pH sampling d | Adjusted pH e

- Re-standardization of the benchtop pH meter is needed at Step 14.12 when the pH is ≥ 4.0.
- Record the standardization of the benchtop pH meter with pH 4.0 and pH 10.0 standards.

  pH Meter ID

§7 Initial Product and UF Membrane Information CONC / DIAFILTRATION

- Record the room number where this process will occur.

  Room Number

§8 WFI Flush prior to Equilibration CONC / DIAFILTRATION

- Confirm that the pH meter has been standardized per SOP-0080297. Confirm that the conductivity meter has been standardized per SOP-0080297.

  Meter ID

§16 Recovery CONC / DIAFILTRATION

- Obtain the tare weight for the Dilution Vessel. Record the equipment ID and calibration due date for the scale and the tare weight of the bag. Check if the Mixer Drive is attached when the tare weight is obtained.

  Scale/Balance ID | Scale/Balance Calibration Due Date | Tare Weight of Dilution Vessel before attachment | Mixer Drive attached at Tare Weight? (check one) | □ Yes | □ No

§17 Dilution CONC / DIAFILTRATION

- Once the Pre-Formulated Intermediate has confirmed to be in range, begin to mix the product per SOP-0107097 by agitation. If using a stir plate, record the equipment ID for the stir plate used. NOTE: Agitation may already be on.

  Mixer/Stir Plate ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥ 15 minutes)

§18 Filtration and Storage of the Pre-Formulated Intermediate CONC / DIAFILTRATION

- Obtain a tare weight of the PFI storage vessel. Record the equipment ID and calibration due date for the scale.

  Scale/Balance ID | Scale/Balance Calibration Due Date | Tare Weight of Storage Vessel

§19 Post-Use WFI Flush CONC / DIAFILTRATION

- Confirm that the conductivity meter has been standardized per SOP-0080297.

  Meter ID

Instantiated wherever the process needs it, so the text is collected from across both records (deduped); the same lines also appear under their unit-op card.

SMPL: Additional Manufacturing Supplies NEW

Original BR: §4

Instructions

If additional materials (filter, bag, assembly, etc.) are used during processing, verify the material is acceptable per Section 4 and record its information; document the reason in the Comments Section. Enter the Part No. (Material Description auto-fills) and Batch No. (Exp. Date auto-fills). **SAP consumption posts automatically via the Goods Issue process message (Z_PICONS, movement type 261).** Built new: a variant of SMPL: Additional Assembly with an added MPR Step No.

Reuses: **Variant of DE1 100: SMPL: Additional Assembly (adds MPR Step No. / Material Description) — no known client variant**

Data Input

#	MPR Step No.	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Autoclave ID	Autoclave Exp.	Performed By *
1								⏱	

+ Add Row

Witnessed By *

SMPL: Additional Solution Batches (Solution Switch) NEW

Original BR: §5

Instructions

When a process solution must be replaced with a different batch during processing, record the switch: indicate the applicable step, confirm the new solution is the same Part No. as the one replaced, and record Part No. / Batch No. / Exp. Date. Document the reason in the Comments Section. Built new: a variant of SMPL: Material Consumption (which carries Part No. / Batch / Exp.) with an added Applicable Step No.

Reuses: **Variant of DE1 100: SMPL: Material Consumption (adds Applicable Step No.; field-verified NOT Solution Summary) — no known client variant**

Data Input

#	Applicable Step No.	Part Number	Batch Number	Exp. Date	Performed By *
1					

+ Add Row

Witnessed By *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 13 GROUP(S)):

§5 Document Process Flow Sheet VIRUS INACTIVATION

- (records)

Starting Material | PN: 8012440: AZD0543 Affinity Product | Acid Addition | PN: A-005168: 1M Acetic Acid | VI Incubation (Target 75 minutes; NOR 60-240 minutes) | Base Addition | PN: A0101-D: 1M Tris Base | Recorded by / date

- Additional Solution Batches Used in Process
- If a solution (of a different batch noted in Section 5) needs to be replaced during processing:
- Indicate the switch at the step it is occurring. Ensure the new solution is the same part number as the solution being replaced.
- Record the solution part number, batch number (and/or process order number) and expiration date below.
- Document the change in Attachment 7: Comments Section.

§5 Process Flowchart CONC / DIAFILTRATION

- (records)

WFI Flush prior to Equilibration | Water for Injection (WFI) | Equilibration | 8012555: 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 | Concentration 1 of the Virus Filtration (VF) Product (8012474) to 100 g/L | Diafiltration | Concentration 2 to 210 g/L | Recovery flush to 190 g/L | Dilution to 180 g/L | Filtration, Sampling and Storage | System Sanitization | A0013-D: 0.5 N Sodium Hydroxide

- Sanitization buffer batch number and exp date are documented in Step 20.3.

System Storage | A0014-D: 0.1 Sodium Hydroxide | Storage buffer batch number and exp. date are documented in Step 20.3. | Recorded by/ Date

- Additional Solution Batches Used in Process
- If a solution (of a different batch noted in Section 5) needs to be replaced during processing:
- Indicate the switch at the step it is occurring. Ensure the new solution is the same part number as the solution being replaced.
- Record the solution part number, batch number (and/or process order number) and expiration date below.
- Document the change in Attachment 9: Comments Section.

Applicable Step Number | Part Number | Batch Number

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Process Notes & Global Limits (Acknowledgement) REUSE

Original BR: §6

Global process notes / limits (15-27 °C; 1 L = 1 kg; within-expiry; equipment-ID attests calibration; Targets/Alert Limits; NaOH corrosive). The operator acknowledges; each limit is enforced at the step that captures its parameter. Reused as-is (Text Instructions shell, content per MBR).

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (notes authored in IV_INSTR; DE2 903: Long Text Instructions)**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 16 GROUP(S)):

§6 Process Notes VIRUS INACTIVATION

- All activities are performed at 15 □C to 27 □C.
- For all calculations, assume 1 L = 1 kg and 1 mL = 1 g unless otherwise noted.
- Verify that all solutions, raw materials, and standards are within expiration prior to use.
- When an equipment ID is written in the MPR, the operator is indicating they have visually inspected and verified calibration prior to use.
- If additional mixing/recirculation is required after initial sampling of the product, record the mixing information in Attachment 6: Product Recirculation Worksheet.
- If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.
- If Non-Routine samples are taken, re-adjust the weight of the product sampled.
- All parameters listed in this document are Targets/Alert Limits unless otherwise noted. When alert limits are exceeded, contact Supervisor and Technical Transfer Group. Document any actions taken in the Attachment 7: Comments Section of the MPR.

§6 Process Notes CONC / DIAFILTRATION

- All activities are performed at 15 □C to 27 □C.
- For all calculations, assume 1 L = 1 kg and 1 mL = 1 g unless otherwise noted.
- Verify that all solutions, raw materials, and standards are within expiration prior to use.
- When an equipment ID is written in the MPR, the operator is indicating they have visually inspected and verified calibration prior to use.
- If additional mixing/recirculation is required after initial sampling of the product, record the mixing information in Attachment 9: Comments Section.
- CAUTION: Sodium Hydroxide is corrosive. A face shield must be worn when the solution is being connected and/or disconnected from the UF/DF system.
- If additional WFI is pulled from the WFI drop during processing, record the WFI valve ID in Attachment 9: Comments Section.
- If a non-routine sample is requested by memo, complete Attachment 4: Non-Routine Sampling Record.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

Reusable Building Blocks (instantiated across BOTH records)

SMPL: Incoming Product Information (Weight / Concentration / DLIMS) NEW

Original BR: VI 7.5 · C&D 7.4-7.6

Instructions

Record the incoming / starting product for each vessel or bag: **tare + net weight** (kg = L), the **SoloVPE protein concentration** (g/L, sample per SOP-0107091) and the **DLIMS project / sample numbers** it is reported under. Built new — no existing block combines these. **Goods issue has no batch field → consume the upstream Affinity / Pod-Harvest PN/batch.**

Reuses: **NEW — no DE1 100 block combines weight + concentration + DLIMS (field-verified: Material Consumption has PN/Batch, Solution Summary has measurement, neither has both)**

Data Input

Vessel / Bag #	Tare Weight (kg=L)	Net Weight / Volume (kg=L)	SoloVPE Concentration (g/L)	DLIMS Project No.	DLIMS Sample No.	Performed By *
1						

+ Add Row

Witnessed By *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 17 GROUP(S)):

§7 Affinity Product Information / Pre-Processing Activities VIRUS INACTIVATION

- Record the room where this process will occur.

  Room Number

- If the Affinity Product was warmed up (from cold storage), ensure the time the product was removed from cold storage is recorded in Attachment 3: Hold Times.
- Confirm that the pH meter has been standardized between pH 2.0 and pH 7.0 per SOP-0080297. Confirm that the conductivity meter has been standardized within the acceptable range per SOP-0080297.

  Meter ID

- Confirm that the thermometer is within its calibration interval. Record the equipment ID of the thermometer used.

  Thermometer ID

- Record the tare weight and net weight of the AZD0543 – Affinity Product vessel.

  Tare Weight of the Affinity Product Vessel | kg = L | Net Weight of the Affinity Product | kg = L | Affinity Product SoloVPE Concentration (8012440-30) | DLIMS Project Number | DLIMS Sample Number

- Mix the product for at least 15 minutes. Follow SOP-0080423 for the LevTech mixer, if applicable. Follow SOP-0107102 for the Magnetic Mixer, if applicable. Follow SOP-0107095 for fixed tanks, if applicable. NOTE: If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.

  Mixer ID/Tank ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥ 15 minutes)

- Remove at least the sample volume per Attachment 1: Sampling Record and SOP-0107097. If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.

  Sampling Date and Time | Date: | Time:

- Aliquot the sample (8012441-10) per Attachment 1: Sampling Record. If applicable, aliquot non-routine samples per Attachment 2: Non-Routine Sampling Record. Label the sample per SOP-0107056.
- Submit the sample to the storage location indicated in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.
- Measure the pH, conductivity, and temperature of the Affinity product per SOP-0080297 and SOP-0107117. Record the results below.

  Conductivity | mS/cm | Temperature

- Obtain the gross weight of the Affinity product vessel after sampling. NOTE: Ensure the vessel is in the same conformation as when the tare weight was obtained.

  Scale/Balance ID | Calibration Due Date | Gross Weight/Volume of the Affinity Product vessel after sampling | kg=L

- Calculate the net weight of the Affinity product after sampling:

  Gross Weight/Volume of the Affinity Product vessel after sampling (Step 7.11) | Tare Weight of the Affinity Product Vessel (Step 7.5) | Affinity Weight/Volume after sampling | kg=L

- If the product will be transferred to a different adjustment vessel, N/A Step 7.14 and go to Section 8. Otherwise: N/A Section 8 and re-label the Viral Inactivation (VI) vessel “VI Treatment Vessel” and record the tare weight on the label. Continue to Step 7.14.

  Starting Product Temperature Check Proceed per the table below. | Check One | Temperature of the sample at Step 7.10 is: | Action | Within 18 – 25 oC

- Transcribe the pH, conductivity, and temperature results into Step 10.9 “Initial pH / Conductivity / Temperature.” N/A all of Section 9 and continue to Step 10.1.

  Less than 18 oC:

- Turn on the treatment vessel’s agitator and document the mixing start time in Step 9.1. * Begin warming the vessel per SOP-0080503/SOP-0080023 (for a bag) or SOP-0107095 (for a tank) to 18 – 25 oC. ** Continue to Section 9.

  Greater than 25 oC

- Contact a supervisor and document actions taken in Attachment 7: Comments Section.
- * NOTE: The product warming range is 18 – 25 oC. ** NOTE: If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Product Vessel Weigh REUSE

Original BR: VI §7/8/15 · C&D §9/11/16/17/18

Get Scale / Balance (ID + Calibration Due auto-populate), record gross weight; net computed (Gross − Tare, 1 kg = 1 L). Reused as-is.

Reuses: **DE1 100: SMPL: Record Scale Weight (Tare Weight / Select Vessel Type & Tare Weight variants)**

▼ OLD PAPER BATCH RECORD — VERBATIM TEXT THIS REUSABLE XSTEP MUST FULFILL, HARVESTED FROM EVERY SECTION THAT RECORDS IT (13 SECTION(S), 20 GROUP(S)):

§7 Affinity Product Information / Pre-Processing Activities VIRUS INACTIVATION

- Record the tare weight and net weight of the AZD0543 – Affinity Product vessel.

Tare Weight of the Affinity Product Vessel | kg = L | Net Weight of the Affinity Product | kg = L | Affinity Product SoloVPE Concentration (8012440-30) | DLIMS Project Number | DLIMS Sample Number

- Obtain the gross weight of the Affinity product vessel after sampling. NOTE: Ensure the vessel is in the same conformation as when the tare weight was obtained.

Scale/Balance ID | Calibration Due Date | Gross Weight/Volume of the Affinity Product vessel after sampling | kg=L

- If the product will be transferred to a different adjustment vessel, N/A Step 7.14 and go to Section 8. Otherwise: N/A Section 8 and re-label the Viral Inactivation (VI) vessel "VI Treatment Vessel" and record the tare weight on the label. Continue to Step 7.14.

Starting Product Temperature Check Proceed per the table below. | Check One | Temperature of the sample at Step 7.10 is: | Action | Within 18 – 25 oC

§8 Transfer Vessel Information and Set up VIRUS INACTIVATION

- For Step 8.4: VI Treatment Vessel Information: Obtain the tare weight of the Treatment Vessel and record the the information in Step 8.4.

VI Treatment Vessel | Bag Information (if applicable) | Bag Information | Mark the applicable attachments connected to the mixer bag during the vessel tare weight | Attached: | Mark the applicable attachments connected to the mixer bag during the vessel tare weight | Mixer Drive | Top Base | Magnetic Clamp | Tank Information (if applicable) | Tank ID | CIP Batch ID | CIP Exp. Date / Time | Date: | Time: | Pressure Test ID | Pressure Test Date | SIP Batch ID | Filter Information (See Section 4 for acceptable filters and autoclaving requirements.) | Vent Filter (if applicable) | Product Filter 1 | Product Filter 2 | Product Filter 3 | Tare Weight Information | * (NOTE: If collection vessel is a bag, tare weight will be obtained prior to attachment of the product filter.) | Scale/ Balance ID | Calibration Due Date | Tare Weight of the VI Treatment Vessel * | Mark filters attached when tare weight is obtained. | Product filter | Vent filter | No filters, vessel is a bag | Recorded by / Date

- Obtain the gross and net weight of the Affinity product after sampling.

Scale/Balance ID | Calibration Due Date | Gross Weight/Volume of the Affinity Product after sampling | Net Weight/Volume of the Affinity Product after sampling * | * Net Weight/Volume of the Affinity Product after sampling = Gross Weight/Volume of the Affinity Product after sampling (above) – Tare weight of the Product Vessel (Step 8.4) | Starting Product Temperature Check Proceed per the table below. | Check One | Temperature of the sample at Step 8.9 is: | Action | Within 18 – 25 oC

§10 Setup and Acid Addition VIRUS INACTIVATION

- Record the gross weight of the VI Treatment Vessel after priming and the scale/balance used in Step 10.9.

§14 Setup and Base Addition VIRUS INACTIVATION

- Record the gross weight of the VI Treatment Vessel after priming in Step 14.11.
- Obtain and record the gross weight of the VI Treatment Vessel after the neutralization. NOTE: Ensure the VI Treatment Vessel is in the same configuration as when the tare weight was obtained in Step 7.5 (or Step 8.4 if product was transferred to a treatment vessel).

Scale/Balance ID | Calibration Due Date | Gross Weight after Neutralization

§15 VIral-Inactivated Product – Sampling VIRUS INACTIVATION

- Obtain the gross and net weights of the collected product after sampling and record the values.

Scale/Balance ID | Calibration Due Date | Gross Weight of the VI Treatment Vessel after sampling

§7 Initial Product and UF Membrane Information CONC / DIAFILTRATION

- In the Step 7.6 "Virus Filtration Product Information" table: If a tank was used, N/A Vessels 2 and 3. For bag(s), N/A any unused Vessel fields. Record the net weight of each Virus Filtration Product from (MABR-0027575) MPR or the label of each product transport vessel. Record the tare weight of each Virus Filtration Product Vessel. Record the product concentration of each Virus Filtration Product Vessel. Record the corresponding DLIMS project number and sample number for the Virus Filtration Product(s) vessels.

Virus Filtration Product Information | Vessel 1 | Vessel 2, if applicable | Vessel 3, if applicable | Virus Filtration Product Net Weight | Tare Weight | VF Product Concentration | DLIMS Sample Number | DLIMS Project Number | Recorded By/Date

§9 End of Virus Filtration Product Storage CONC / DIAFILTRATION

- Weigh the product vessel(s) and record below. N/A any unused Vessel fields. Calculate the Net Weight for each vessel. The "Total Net Weight after sampling" is the sum of the net weights for each vessel.

Vessel | 2, if applicable | 3, if applicable | Gross Weight | Virus Filtration Product Vessel Tare Weight (Step 7.6) | Net Weight a | Total Net Weight of VF Product after sampling (Vessel 1 + Vessel 2 + Vessel 3) | Scale/Balance ID | Cal. Due Date | Net Weight = Gross weight – Virus Filtration Product Vessel Tare Weight

§11 Vessel Set-up CONC / DIAFILTRATION

- Record the tare weight of the vessel before it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.

Scale / Balance ID | Scale/Balance Calibration Due Date | Tare weight of Retentate Vessel before attachment

- Record the tare weight of the vessel after it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.

Scale / Balance ID | Scale / Balance Calibration Due Date | Tare weight of Retentate Vessel after attachment | Secure the wheels of the Retentate Vessel. | Tare (zero) the scale holding the Retentate Vessel.

§13 Concentration 1 CONC / DIAFILTRATION

- Record the net weight of the product at the end of Concentration 1 from the retentate scale and scale information.

Net Weight of Product after Concentration 1 | Scale ID | Cal. Due Date

§14 Diafiltration CONC / DIAFILTRATION

- Record the net weight of the product at the end of Diafiltration from the retentate scale and scale information.

Net Weight of Product after Diafiltration | Scale ID | Cal. Due Date

§16 Recovery CONC / DIAFILTRATION

- Obtain the tare weight for the Dilution Vessel. Record the equipment ID and calibration due date for the scale and the tare weight of the bag. Check if the Mixer Drive is attached when the tare weight is obtained.

Scale/Balance ID | Scale/Balance Calibration Due Date | Tare Weight of Dilution Vessel before attachment | Mixer Drive attached at Tare Weight? (check one) | ☐ Yes | ☐ No

- Record the gross weight of the Dilution Vessel after transfer. Record the floor scale/balance ID number and calibration due date.

  Gross weight of the Dilution Vessel | Scale/Balance ID | Scale/Balance Calibration Due Date

§17 Dilution CONC / DIAFILTRATION

- Disconnect the buffer line from the Dilution Vessel, if applicable. Remove the dilution vessel from the scale and tare (zero) the scale. Obtain the gross weight of the Dilution Vessel after dilution. NOTE: Ensure the Dilution Vessel is in the same configuration as it was when its tare weight was obtained in Step 16.20.

  Gross weight of Dilution Vessel, with attachments removed | Scale/Balance ID | Calibration Due Date

- Record the gross weight of the Dilution Vessel after sampling. NOTE: Ensure the vessel is in the same configuration as it was when the tare weight was obtained in Step 16.20.

  Dilution Vessel Gross Weight after sampling

§18 Filtration and Storage of the Pre-Formulated Intermediate CONC / DIAFILTRATION

- Record the net weight of the PFI (Step 18.8), the tare weight of the PFI storage vessel (Step 18.2), and the gross weight of the PFI storage vessel after transfer (Step 18.7) on the storage vessel label.

Instantiated wherever the process needs it, so the text is collected from across both records (deduped); the same lines also appear under their unit-op card.

SMPL: Calc — Three Columns (Value1 op Value2 = Result) REUSE

Original BR: VI 7.12 / 8.10 / 14.15 · C&D §7/16/17

Two-operand calc archetype (Value 1 [× ÷ + −] Value 2 = Result) — e.g. VI 7.12 Net = Gross − Tare. Reused as-is.

Reuses: **DE1 100: SMPL: Calc Three Columns (2-input — field-verified: Input Val1 / Input Val2 / Result)**

▼ OLD PAPER BATCH RECORD — VERBATIM TEXT THIS REUSABLE XSTEP MUST FULFILL, HARVESTED FROM EVERY SECTION THAT RECORDS IT (15 SECTION(S), 46 GROUP(S)):

§7 Affinity Product Information / Pre-Processing Activities VIRUS INACTIVATION

- Calculate the net weight of the Affinity product after sampling:

Gross Weight/Volume of the Affinity Product vessel after sampling (Step 7.11) | Tare Weight of the Affinity Product Vessel (Step 7.5) | Affinity Weight/Volume after sampling | kg=L

§9 Additional Mixing and Pre-Treatment Product Warming (if applicable) VIRUS INACTIVATION

- Record the mixing end time. Calculate the mixing duration from the time the mixing was started in Step 9.1.

Mixing End Time | Mixing Duration (≥15 minutes)

§10 Setup and Acid Addition VIRUS INACTIVATION

- Calculate the expected amount of acid to be added based on the addition ratio. * Use Step 8.10 if the product was transferred.

Net Weight of Affinity after sampling (Step 7.12 or 8.10 *) | Acid Addition Ratio (kg acid/ kg Affinity Product) | Recommended weight of acid

- Determine the initial acid addition weight. After initial addition, more acid may be added until the target pH is reached.

Recommended weight of acid (Step 10.6) | Initial Addition Ratio | Target Weight for Initial Acid Addtion

§11 Incubation VIRUS INACTIVATION

- Calculate the duration that the product has been incubating.

Time of Sampling (Step 11.2) | Start Time for Incubation (pH and temperature in spec) (Step 10.9) | Duration the product has been incubating

§14 Setup and Base Addition VIRUS INACTIVATION

- Calculate the expected amount of base to be added based on the addition ratio.

Total Net Weight of Acid added (Step 10.9 or 12.4 *) | Base Addition Ratio (kg base / kg acid) | Recommended weight of base

- Determine the initial base addition weight. After initial addition, more base may be added to reach the target pH.

Recommended weight of base (Step 14.6) | Initial Addition Ratio | Target Weight of Initial Base Addition

- For Step 14.11 - End of Incubation Information: Take a final sample of the incubation product and measure the pH and temperature. In Step 14.11: Record the (A) 'pH at the end time of Incubation' and (B) "Temperature at the end of Incubation' Record the (C) 'End time for Incubation', which is also the start time of the first base addition. Calculate the (D) 'Total Duration of Incubation' NOTE: Contact a supervisor if the Total Duration of Incubation exceeds 240 minutes and document actions taken in Attachment 7: Comments Section.
- Calculate the net weight of the Neutralized VI product. * Use Step 8.10 if the Affinity Product was transferred to a treatment vessel. ** Use Step 12.4 if additional acid was needed. Ensure that the acid from the first addition is included in the total net weight of acid added.

Net Weight of Affinity Product after sampling (Step 7.12 or 8.10 *) | Total Net Weight of Acid added (Step 10.9 or 12.4 **) | Total Net Weight of Base added (Step 14.11) | Net Weight/Volume of the Neutralized VI Product | kg=L

§15 VIral-Inactivated Product – Sampling VIRUS INACTIVATION

- Calculate the net weight/volume of the VI Product after sampling. * NOTE: Use Step 8.4 as tare weight if the Affinity product was transferred to a treatment vessel in Section 8. Otherwise, use Step 7.5 as tare weight.

Gross Weight of the VI Treatment Vessel after sampling (Step 15.6) | Tare Weight of the VI Product Vessel (Step 7.5 or 8.4)* | Net Weight/Volume of the VI Product after sampling | kg = L

§19 Attachment 3: Hold Times VIRUS INACTIVATION

- Calculate the storage duration for each temperature where the product was stored as well as the combined storage duration, if applicable.

Hold | Location | Temp. | Affinity Product Storage Start (Time/Date) | Affinity Product Storage End (Time/Date) | Hold Times | Hold | Location | Temp. | Affinity Product Storage Start (Time/Date) | Affinity Product Storage End (Time/Date) | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | Total R. T. Hold Time: If R.T. Total Hold Time > 72 hrs, contact a supervisor | Total 2-8 °C Hold Time | Total 2-8 °C + RT Hold Time: If Total Combined Hold > 168 hrs, contact a supervisor

§7 Initial Product and UF Membrane Information CONC / DIAFILTRATION

- Calculate the Virus Filtration Product volume in each vessel. NOTE: N/A any unused Vessel fields below.

Virus Filtration Product Net Weight (Step 7.6) | Density of VF Product | Virus Filtration (VF) Product Volume | Vessel 1 | 1.014 kg/L | Vessel 1 | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L

- Calculate the total grams of Virus Filtration Product in each vessel. NOTE: N/A any unused Vessel fields below.

Virus Filtration (VF) Product Volume (Step 7.7) | Virus Filtration Product Concentration (Step 7.6) | Total grams of VF Product per Vessel | Vessel 1 | Vessel 1 | Vessel 1 | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Total grams of VF Product before sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)

- Calculate the total membrane surface area needed for the UFDF process:

Total grams of VF Product before sampling (Step 7.8) | Maximum Load (g of Product / m2) | Minimum membrane surface area

§8 WFI Flush prior to Equilibration CONC / DIAFILTRATION

- Calculate the target volume of solution needed for a 22.0 L/m2 WFI and Equilibration flushes.

Target Flush Volume | Total Surface Area of Cassettes (Step 7.2) | Target WFI and Equilibration Flush Volume

§9 End of Virus Filtration Product Storage CONC / DIAFILTRATION

- Weigh the product vessel(s) and record below. N/A any unused Vessel fields. Calculate the Net Weight for each vessel. The "Total Net Weight after sampling" is the sum of the net weights for each vessel.

Vessel | 2, if applicable | 3, if applicable | Gross Weight | Virus Filtration Product Vessel Tare Weight (Step 7.6) | Net Weight a | Total Net Weight of VF Product after sampling (Vessel 1 + Vessel 2 + Vessel 3) | Scale/Balance ID | Cal. Due Date | Net Weight = Gross weight – Virus Filtration Product Vessel Tare Weight

- Calculate the Virus Filtration Product Volume after sampling. NOTE: N/A any unused Vessel fields below.

Net Weight (Step 9.8) | Density of VF Product | Virus Filtration Product Volume after sampling | Vessel 1 | 1.014 kg/L | Vessel 1 | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L

- Calculate the Total Grams of Virus Filtration Product after sampling. NOTE: N/A any unused Vessel fields below.

Virus Filtration Product Volume after sampling (Step 9.9) | Virus Filtration Product concentration (Step 7.6) | Total Grams of Virus Filtration Product after sampling per Vessel | Vessel 1 | Vessel 1 | Vessel 1 | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Total grams of Virus Filtration Product after sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)

§12 Process Input Calculations CONC / DIAFILTRATION

- Calculate the volume at 80% Retentate Vessel Capacity.

Retentate Vessel Volume (Step 11.1) | Capacity Limit | Retentate Vessel Charge Volume/Weight at 80% Capacity | L = kg

- Calculate the volume at 40% Retentate Vessel Capacity.

Retentate Vessel Volume (Step 11.1) | Buffer Charge Target | Retentate Vessel Charge Volume/Weight at 40% Capacity | L = kg

- Calculate the product volume in the UF/DF Skid after Concentration 1.

Total Grams of VF Product after sampling (Step 9.10) | Concentration 1 Range | Product volume in the UF/DF System after Concentration 1 | 110 g/L (max) | L (Min) | 100 g/L (Target) | L (Target) | 90 g/L (min) | L (Max)

- Calculate the product volume in the Retentate Vessel after Concentration 1.

Product volume in the UF/DF System after Concentration 1 (Step 12.3) | Hold-up volume of the UF/DF Skid (Step 7.2) | Product volume in the Retentate Vessel after Concentration 1 | L (Min) | L (Min) | L (Target) | L (Target) | L (Max) | L (Max)

- Calculate the product weight in the Retentate Vessel after Concentration 1.

Product volume in the Retentate Vessel after Concentration 1 (Step 12.4) | Density of the Concentration 1 Product | Product weight in the Retentate Vessel after Concentration 1 | L (Min) | kg/L | kg (Min) | L (Target) | kg/L | kg (Target) | L (Max) | kg/L | kg (Max)

- Calculate the permeate volume range for Diafiltration.

Diafiltration Volume Range (DVs) | Product Volume in the UF/DF System after Concentration 1 (Step 12.3) | Permeate volume range after Diafiltration | L (Min) | L (Target)

- Calculate the product volume range in the UF/DF System after Concentration 2

Total Grams of VF Product after sampling (Step 9.10) | Concentration 2 Range | Product volume in the UF/DF System after Concentration 2 | 210 g/L (target) | L (Target) | g/L (min) | L (Max)

- Calculate the product volume range in the Retentate Vessel after Concentration 2.

Product volume in the UF/DF System after Concentration 2 (Step 12.7) | Hold-up volume of the UF/DF Skid (Step 7.2) | Product volume in the Retentate Vessel after Concentration 2 | L (Min) | L (Min) | L (Target) | L (Target) | L (Max) | L (Max)

- Calculate the net weight range of the product in the Retentate Vessel after Concentration 2.

Product volume in the Retentate Vessel after Concentration 2 (Step 12.8) | Density of the product after Concentration 2 at target | Net Weight of product in the Retentate Vessel after Concentration 2 | L (Min) | kg/L | kg (Min) | L (Target) | kg/L | kg (Target) | L (Max) | kg/L | kg (Max)

§14 Diafiltration CONC / DIAFILTRATION

- Calculate the product volume in the Retentate Vessel after Diafiltration.

Net Weight of Product after Diafiltration (Step 14.8) | Density of the product after Diafiltration | Product Volume in the Retentate Vessel after Diafiltration | kg/L

- Calculate the product volume in the Retentate Vessel after the additional Diafiltration.

Net Weight of Product after Diafiltration (Step 14.19) | Estimated Density of the Product after Diafiltration | Product Volume in the Retentate Vessel after Diafiltration | kg/L

§16 Recovery CONC / DIAFILTRATION

- Calculate the Recovery Product volume range.

Total grams of Virus Filtration Product after sampling (Step 9.10) | Recovery Concentration Range | Recovery Product Volume Range | L (Target) | L (Max)

- Calculate the Recovery Product weight range.

Recovery Product Volume Range (Step 16.1) | Density of the Product at Recovery | Recovery Product Net Weight Range | L (Target) | 1.065 kg/L | kg (Target) | L (Max) | 1.065 kg/L | kg (Max)

- Calculate the amount of Recovery buffer to add.

Recovery Product Net Weight range (Step 16.2) | Retentate Weight after Recirculation (Step 15.10) | Weight of Recovery buffer to add | kg (Target) | kg (Target) | kg (Max) | kg (Max)

- Calculate the volume of Recovery buffer to add.

Target Weight of Recovery buffer to add (Step 16.3) | Density of the Recovery buffer | Volume of Recovery buffer to add | 1.008 kg/L

- Calculate the amount of Recovery buffer 8012555 used for the Recovery flush.

Retentate Vessel Weight after Recovery (with attachments) (Step 16.14) | Retentate Weight after Recirculation (Step 15.10) | Amount of Recovery buffer 8012555 used for Recovery flush

- Calculate the net weight of the Recovery Product after transfer.

Gross weight of the Dilution Vessel (Step 16.31) | Tare weight of Dilution Vessel before attachment (Step 16.20) | Net weight of Recovery Product after Transfer

- Calculate the volume of the Recovery Product after Transfer.

Net weight of Recovery Product after Transfer (Step 16.32) | Density of the product at 190 g/L | Volume of Recovery Product after Transfer | kg/L

§17 Dilution CONC / DIAFILTRATION

- Calculate the total grams of product in the Retentate Vessel.

Volume of Recovery Product after Transfer (Step 16.33) | Concentration of the Recovery Product (Step 16.34) | Total grams of the Recovery Product

- Calculate the AZD0543 Dilution Product Volume Range.

Total grams of the Recovery Product (Step 17.1) | PFI Concentration Range | AZD0543 Dilution Product Volume Range | g/L (max) | L (Min) | 180 g/L (target) | L (Target) | g/L (min) | L (Max)

- Calculate the AZD0543 Dilution Product Net Weight Range.

AZD0543 Dilution Product Volume Range (Step 17.2) | Density of the Pre-Formulated Intermediate (PFI) | AZD0543 PFI Net Weight Range | L (Min) | 1.062 kg/L | kg (Min) | L (Target) | 1.062 kg/L | kg (Target) | L (Max) | 1.062 kg/L | kg (Max)

- Calculate the weight of Dilution Buffer to Add.

AZD0543 PFI Net Weight Range (Step 17.3) | Net weight of the Recovery Product after Transfer (Step 16.32) | Weight of Dilution Buffer to Add | kg (Min) | kg (Min) | kg (Target) | kg (Target) | kg (Max) | kg (Max) | Convert the Weight of Dilution Buffer to Add to volume. | Weight of Dilution Buffer to Add (Step 17.4) | Density of the Dilution Buffer (PN: 8012555) | Volume of Dilution Buffer to Add | kg (Min) | 1.008 kg/L | L (Min) | kg (Target) | 1.008 kg/L | L (Target) | kg (Max) | 1.008 kg/L | L (Max)

- Calculate the net weight of PFI.

Gross weight of Dilution Vessel, with attachments removed (Step 17.13) | Tare weight of Dilution Vessel before attachment (Step 16.20) | PFI Net Weight

- Calculate the volume of PFI after dilution and before sampling.

PFI Net Weight (Step 17.14) | Density of the PFI | PFI Volume before sampling | 1.062 kg/L

- Calculate the net weight of the Pre-Formulated Intermediate (PFI) after sampling.

Dilution Vessel Gross Weight after sampling (Step 17.30) | Tare weight of Dilution Vessel (Step 16.20) | PFI Net Weight after sampling

- Calculate the volume of the PFI after sampling.

PFI Net Weight after sampling (Step 17.31) | Density of the PFI | PFI Volume after sampling | 1.062 kg/L

§18 Filtration and Storage of the Pre-Formulated Intermediate CONC / DIAFILTRATION

- Calculate the net weight of the PFI after sampling and transfer

Gross Weight of PFI Storage Vessel after transfer (Step 18.7) | Tare Weight of Storage Vessel (Step 18.2) | Net Weight of the PFI after sampling and transfer

- Calculate the volume of the PFI after sampling and transfer.

Net Weight of the PFI after sampling and transfer (Step 18.8) | Density of the Product after Dilution | Volume of the PFI after sampling and transfer | 1.062 kg/L

Instantiated wherever the process needs it, so the text is collected from across both records (deduped); the same lines also appear under their unit-op card.

SMPL: Multi-Variable Calculation (3-input / ranged) REUSE

Original BR: C&D §12/16/17

Multi-operand calc archetype — 3-input (Three Variable Calc) or ranged result (Calc Range Values). Reused as-is.

Reuses: **DE1 100: SMPL: Three Variable Calc (3-input) / SMPL: Calc Range Values (ranged min/target/max)**

SMPL: Product Mixing REUSE

Original BR: VI §15 · C&D §9/16/17

Get Mixer (ID + Description), record agitation RPM, stamp start/end; duration computed. Reused as-is.

Reuses: **DE1 100: SMPL: Mixing Time (field-verified: Agitation RPM + Start/End + Duration of Mixing)**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 10 GROUP(S)):

§9 End of Virus Filtration Product Storage CONC / DIAFILTRATION

- If the Virus Filtration Product was warmed up (from cold storage), ensure the time the product was removed from cold storage is recorded in Attachment 5: Hold Times.
- Mix the Virus Filtration product for at least 15 minutes and record mixing information below: Follow SOP-0080423 for the LevTech mixer. Follow SOP-0107095 for agitation in fixed tanks. If mixing by recirculation, N/A this step and document mixing in Attachment 7: Product Recirculation Work Sheet. N/A any unused Vessel fields below.

  Vessel 1 | Mixer ID (if applicable) | Agitation Speed | Start Time | End Time | Duration (≥ 15 min) | Vessel 2, if applicable | Mixer ID (if applicable) | Agitation Speed | Start Time | End Time | Duration (≥ 15 min) | Vessel 3, if applicable | Mixer ID (if applicable) | Agitation Speed | Start Time | End Time | Duration (≥ 15 min)
- Remove at least the sample volume per Attachment 3: Sampling Record from each of the product collection vessel per SOP-0107097. If non-routine samples are requested by memo, complete Attachment 4: Non-Routine Sampling Record. N/A any unused Vessel fields below.

  Sampling Date and Time, Vessel 1 | Date: | Time: | Sampling Date and Time, Vessel 2, if applicable | Date: | Time: | Sampling Date and Time, Vessel 3, if applicable | Date: | Time:
- Aliquot the sample (sample designation: 8012475-10-1, 8012475-10-2, or 8012475-10-3, as applicable) per Attachment 3: Sampling Record. If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample per SOP-0107056.
- Submit the sample to the storage location per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.
- Place the remaining sample in a 50 ml conical tube and measure the temperature of the sample per SOP-0107117. Record the reading below. N/A any unused Vessel fields below.

  Vessel 1 Temperature | Vessel 2 Temperature, if applicable | Vessel 3 Temperature, if applicable
- If the VF product is in a bag or portable tank, N/A this step. Otherwise, continue below. In Purification, ensure the following: All the necessary transfer panel connections are made for transfer of the VF product to Formulation per SOP-0107099. The condensate flex hose on the transfer panel is removed and the manual valve that the flex hose was attached to is closed. Use SOP-0080548 for Large Purification fixed tanks and SOP-0080323 for Small Purification fixed tanks.
- Weigh the product vessel(s) and record below. N/A any unused Vessel fields. Calculate the Net Weight for each vessel. The “Total Net Weight after sampling” is the sum of the net weights for each vessel.

  Vessel | 2, if applicable | 3, if applicable | Gross Weight | Virus Filtration Product Vessel Tare Weight (Step 7.6) | Net Weight a | Total Net Weight of VF Product after sampling (Vessel 1 + Vessel 2 + Vessel 3) | Scale/Balance ID | Cal. Due Date | Net Weight = Gross weight – Virus Filtration Product Vessel Tare Weight
- Calculate the Virus Filtration Product Volume after sampling. NOTE: N/A any unused Vessel fields below.

  Net Weight (Step 9.8) | Density of VF Product | Virus Filtration Product Volume after sampling | Vessel 1 | 1.014 kg/L | Vessel 1 | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L
- Calculate the Total Grams of Virus Filtration Product after sampling. NOTE: N/A any unused Vessel fields below.

  Virus Filtration Product Volume after sampling (Step 9.9) | Virus Filtration Product concentration (Step 7.6) | Total Grams of Virus Filtration Product after sampling per Vessel | Vessel 1 | Vessel 1 | Vessel 1 | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Total grams of Virus Filtration Product after sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Product Sampling & DLIMS Submission REUSE

Original BR: VI §15/Att1 · C&D §8/14/16/17/Att3

Remove the sample per Att 1/3 & SOP-0107097; aliquot & label per SOP-0107056; submit to DLIMS per SOP-0080295; stamp date/time; record DLIMS project / sample numbers. Reused as-is.

Reuses: **DE1 100: SMPL: Sampling Record + SMPL: Sample Submission Chart**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Sampling Record — pull & aliquot (SOP-0107097 / label SOP-0107056)

Sample Designation *

▸ Record

Sampling Date & Time *

Sample Volume *

Intermediate Storage Temp (°C) *

SMPL: Sample Submission Chart — submit to DLIMS (SOP-0080295)

#	DLIMS Test Name	DLIMS Project No.	DLIMS Sample No.	Tube / Container Type	Location	Submitted?	Performed By *
1						Yes	

+ Add Row

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (3 SECTION(S), 21 GROUP(S)):

§15 Viral-Inactivated Product – Sampling VIRUS INACTIVATION

- Mix the product for at least 15 minutes. Follow SOP-0080423 for the LevTech mixer, if applicable. Follow SOP-0107102 for the Magnetic Mixer, if applicable. Follow SOP-0107095 if using a product hold tank, if applicable. NOTE: If the product is in a non-mixing vessel and requires recirculation to mix, N/A the step below and complete Attachment 6: Product Recirculation Worksheet.

  Mixer ID/Tank ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥ 15 minutes)

- Remove at least the sample volume per Attachment 1: Sampling Record and SOP-0107097. If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.

  Sampling Date and Time | Date: | Time:

- Aliquot the sample (8012441-30) per Attachment 1: Sampling Record. If applicable, aliquot non-routine samples per Attachment 2: Non-Routine Sampling Record. Label the sample per SOP-0107056.
- Submit the sample to the storage location indicated in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.
- Measure the pH, conductivity, and temperature of the sample. NOTE: Contact Supervisor if the pH of the VI Product is not 7.2 – 7.6 and document actions taken in Attachment 7: Comments Section.

  pH (7.2 – 7.6) | Conductivity | mS/cm | Temperature

- Obtain the gross and net weights of the collected product after sampling and record the values.

  Scale/Balance ID | Calibration Due Date | Gross Weight of the VI Treatment Vessel after sampling

- Calculate the net weight/volume of the VI Product after sampling. * NOTE: Use Step 8.4 as tare weight if the Affinity product was transferred to a treatment vessel in Section 8. Otherwise, use Step 7.5 as tare weight.

  Gross Weight of the VI Treatment Vessel after sampling (Step 15.6) | Tare Weight of the VI Product Vessel (Step 7.5 or 8.4)* | Net Weight/Volume of the VI Product after sampling | kg = L

- Store the product and document storage information below. For product stored at 2 – 8ºC, the storage start time is the time when the product is transferred into 2 – 8ºC storage. For product stored at 15 – 25ºC, the storage start time is the time when final sampling occurred in Step 15.2.
- Ensure that Attachment 3: Hold Times is filled out with all hold times for the Affinity product.

  Name in MPR | Name in Attachment | Step No. | Start time of addition | Affinity Product Storage End (Time/Date) | Step 10.9

§17 Attachment 1: Sampling Record VIRUS INACTIVATION

- (records)

  Viral Inactivation | Dev Samples DLIMS Project # | Pre-Load Performed By/Date | DLIMS Test Name | Volume | Location | Tube Type | Intermediate Storage Temp b | DLIMS Storage Temp | Sample Number | -10 (Affinity Product) | BIOBURDEN_IP | ENDOTOXIN_IP_GPF1 | PET | -30 (VI Product) | SOLO_A280 a | CIEF_NATIV | CHO_SYBR | EL_CHO_GEN | SEC_PUR | CGE_N_RED | ABT Retain | 5 x 14 mL | PPS Small Retain | 3 x 14 mL

- Test SOLO_A280 sample per SOP-0107091. If sample is not tested immediately, store sample at 2 - 8°C.
- The intermediate storage temperature applies to both the shipment and the storage of the sample prior to reaching the sample management team.
- Use GREENFORM Template for Development DLIMS sample submissions.
- Sample Designation “-10”: For sampling of starting material.
- Sample Designation “-30”: For product quality samples.
- Bold (STAT): Bold text indicates STAT sample submission.

§23 Attachment 3: Sampling Record CONC / DIAFILTRATION

- (records)

  Concentration and Diafiltration | Dev Samples DLIMS Project # | CoA GxP Samples Project # | WHITEFORM GxP Samples Project # | Pre-Load Performed By/Date | DLIMS Test Name | Sample Volume | Location | Tube Type | DLIMS Form | Intermediate Storage Temp (oC) | DLIMS Storage Temp. (oC) | Sample Number | -10 (Virus Filtered Product) | BIOBURDEN_IP_GPF1 | ENDOTOXIN_IP_GPF1 | PET | -20 (Recovered Product) | SOLO_A280 a | - 30 (Diluted Product, if applicable) | SOLO_A280 a | -40 (Pre-Formulated Intermediate - Product Quality Samples) | SEC_PUR | CGE_N_RED | SOLO_A280 a | ABT Retain c | 5 x 14 mL | PPS Small Retain b | 3 x 14 mL | BIOBURDEN_IP_GPF2 | ENDOTOXIN_IP_GPF1 | PET

- Test SOLO_A280 sample per SOP-0107091. If sample not tested immediately, store sample at 2 - 8°C.
- For PPS Small Retains, select “Retain – PPS Small” for the purpose field when loading into DLIMS.
- For AS Retains, select “Retain – ABT” for the purpose field when loading into DLIMS.
- NOTE: Bolded tests are STAT samples.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Product pH / Conductivity / Temperature REUSE

Original BR: VI 7.1/15.5 · C&D 8.12/14.11/17

Get pH/Cond Meter + Get Thermometer, then record range-validated pH / conductivity / temperature per reading. Assembled from two Equipment Selects + the Solution Summary results block (collapses per row). Contact supervisor if out of specification.

Reuses: **DE1 100 stack: SMPL: Equipment Select ×2 (pH/Cond Meter + Thermometer) + SMPL: Solution Summary - Data Recording (range-validated; field-verified = Equipment + Min/Max Target + Actual Value + Cond High/Low)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Description

Calibration Due Date

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Description

Calibration Due Date

SMPL: Solution Summary - Data Recording — range-validated readings (collapses per row)

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High / Low)
1	pH					

+ Add Row

▼ OLD PAPER BATCH RECORD — VERBATIM TEXT THIS REUSABLE XSTEP MUST FULFILL, HARVESTED FROM EVERY SECTION THAT RECORDS IT (9 SECTION(S), 12 GROUP(S)):

§7 Affinity Product Information / Pre-Processing Activities VIRUS INACTIVATION

- Measure the pH, conductivity, and temperature of the Affinity product per SOP-0080297 and SOP-0107117. Record the results below.

Conductivity | mS/cm | Temperature

- Transcribe the pH, conductivity, and temperature results into Step 10.9 “Initial pH / Conductivity / Temperature.” N/A all of Section 9 and continue to Step 10.1.

Less than 18 oC:

§9 Additional Mixing and Pre-Treatment Product Warming (if applicable) VIRUS INACTIVATION

- Measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 “Initial pH / Conductivity / Temperature.” Document the mixing end time in Step 9.4, then continue to Section 10.

Less than 18 oC

- Continue to warm and agitate the product. * Resample the product regularly until the product is within 18 – 25 oC. When the product is within 18 – 25 oC, measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 “Initial pH / Conductivity / Temperature.” Document the mixing end time in Step 9.4, then continue to Section 10.

Greater than 25 oC

§10 Setup and Acid Addition VIRUS INACTIVATION

- Step 10.9: Acid Treatment Begin transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 10.9.

Acid Treatment | Initial Measurements | Initial pH | Initial Conductivity | Initial Temperature | Start time of addition (End of storage time) | Initial Measurements | mS/cm | Scale/Balance ID | Calibration due date | Gross weight of Treatment Vessel before addition (kg) | End time of acid addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) a | Total net weight of buffer added (kg) b | Time of pH sampling | Adjusted pH | Sample Temperature (oC)

§11 Incubation VIRUS INACTIVATION

- Incubated Product Information Confirm that the product has been incubating for a minimum of 60 minutes. Take a 25 mL sample from the incubated product. Measure the pH, conductivity, and temperature of the sample and record the results below.

Time of Sampling | pH (3.4 – 3.6) | Conductivity | mS/cm | Temperature (18 – 25 oC)

§12 Acid Addition- Extension VIRUS INACTIVATION

- Step 12.4: Acid Treatment Extension Continue transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 12.4.

Acid Treatment - Extension | Scale/Balance ID | Calibration Due Date | Start time of addition | Gross weight of Treatment Vessel before addition (kg) | End time of acid addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) a | Total net weight of buffer added (kg) b | Time of pH sampling | Adjusted pH | Sample Temperature (oC) | Time of pH sample confirmed at 3.5 c, d | Final pH (3.4 – 3.6) c | Final Conductivity | ---------------------------- mS/cm | Temperature (18 – 25 oC) e | Start Time for Incubation (pH and Temperature in spec) f

§14 Setup and Base Addition VIRUS INACTIVATION

- For Step 14.11 - End of Incubation Information: Take a final sample of the incubation product and measure the pH and temperature. In Step 14.11: Record the (A) ‘pH at the end time of Incubation’ and (B) “Temperature at the end of Incubation’ Record the (C) ‘End time for Incubation’, which is also the start time of the first base addition. Calculate the (D) ‘Total Duration of Incubation’ NOTE: Contact a supervisor if the Total Duration of Incubation exceeds 240 minutes and document actions taken in Attachment 7: Comments Section.
- For Step Step 14.11 - Neutralization Treatment: After a target incubation time of 75 minutes, within a range of 60 – 240 minutes, begin transferring 1M Tris Base (A0101-D) into the VI Treatment Vessel: Do not stop agitation during treatment. For the base additions, follow SOP-0081015 if the VI Treatment Vessel is a MagMix bag. If the vessel is a tank, follow SOP-0107095. After each addition of A0101-D, mix the product for at least ten minutes prior to taking a sample for pH measurement. When pH rises above 4.0, recalibrate the pH meter to pH 4.0 and pH 10.0 and record the calibration information in Step 14.12. Continue adding A0101-D until the product pool is within pH range 7.2 – 7.6. Record the Final pH, Final conductivity, Temperature and End Time For Incubation in Step 14.11.

Neutralization Information | End of Incubation Information | (A) pH at the end time of Incubation (3.4 – 3.6) | (B) Temperature at the end of Incubation (18 – 25 oC) | (C) End time for Incubation (Start Time of base addition) | Start Time of Incubation (Step 10.9 or 12.4) a | (D) Total Duration of Incubation (60-240 minutes) | minutes | Neutralization Treatment | Scale/Balance ID | Calibration Due Date | Gross weight of Treatment Vessel before addition (kg) | End time of base addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) b | Total net weight of buffer added (kg) c | Time of pH sampling d | Adjusted pH e

§15 VIral-Inactivated Product – Sampling VIRUS INACTIVATION

- Measure the pH, conductivity, and temperature of the sample. NOTE: Contact Supervisor if the pH of the VI Product is not 7.2 – 7.6 and document actions taken in Attachment 7: Comments Section.

pH (7.2 – 7.6) | Conductivity | mS/cm | Temperature

§14 Diafiltration CONC / DIAFILTRATION

- Collect a sample from the permeate and retentate lines for pH, conductivity, and temperature measurement. Record the results below. NOTE: If the Retentate and Permeate pH values are not within range, contact a supervisor prior to proceeding and document actions taken in Attachment 9: Comments Section.

Retentate Values | Conductivity | mS/cm | Temperature (15 – 25°C) | Permeate Values | Conductivity | mS/cm | Temperature (15 – 25°C)

§17 Dilution CONC / DIAFILTRATION

- Aliquot product into a 50 ml conical and measure the pH, conductivity, and temperature. NOTE: If the pH value is not in range, contact a supervisor and document actions taken in Attachment 9: Comments Section.

Conductivity | mS/cm | Temperature (15 – 25 °C)

Instantiated wherever the process needs it, so the text is collected from across both records (deduped); the same lines also appear under their unit-op card.

SMPL: Hold Time Table NEW

Original BR: VI §19 (Att3) · C&D §9/§25 (Att5)

Instructions

Track product hold time across storage moves. Per hold: select the storage condition, stamp start and end; RT and 2-8 °C durations computed separately. Totals computed — alert if RT total exceeds 72 h or the combined total exceeds 168 h (contact supervisor). Built new: no block has the duration + threshold logic.

Reuses: **NEW — no DE1 100 block carries the RT / 2-8 °C hold DURATIONS + >72 h / >168 h threshold logic (Solution Summary is measurement, Solution Final Storage is a location)**

Data Input

Hold #	Location	Storage Temp		Storage Start		Storage End	RT Hold (h)	2-8°C Hold (h)	Performed By *
1		2-8 °C	▸ Record		▸ Record				

+ Add Row

Total RT Hold (h)

Total 2-8°C Hold (h)

Total Combined Hold (h)

Witnessed By *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 6 GROUP(S)):

§19 Attachment 3: Hold Times VIRUS INACTIVATION

- Record the storage start time, end time, and date for the Affinity Product.
- For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage, and the storage end time is the time when the product is transferred from 2 – 8°C storage. A new entry for storage at R. T. (15 – 25°C) begins upon removal from 2 – 8°C.
- Calculate the storage duration for each temperature where the product was stored as well as the combined storage duration, if applicable.

Hold | Location | Temp. | Affinity Product Storage Start (Time/Date) | Affinity Product Storage End (Time/Date) | Hold Times | Hold | Location | Temp. | Affinity Product Storage Start (Time/Date) | Affinity Product Storage End (Time/Date) | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | Total R. T. Hold Time: If R.T. Total Hold Time > 72 hrs, contact a supervisor | Total 2-8 °C Hold Time | Total 2-8 °C + RT Hold Time: If Total Combined Hold > 168 hrs, contact a supervisor

§25 Attachment 5: Hold Times CONC / DIAFILTRATION

- Record the storage start time, end time, and date for the Virus Filtered Product.
- For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage, and the storage end time is the time when the product is transferred from 2 – 8°C storage. A new entry for storage at 15 – 25°C begins upon removal from 2 – 8°C.
- Calculate the storage duration for each temperature where the product was stored as well as the combined storage duration, if applicable.

Hold | Location | Temp. | Virus Filtered Product Storage Start (Time/Date) | Virus Filtered Product Storage End (Time/Date) | Hold Times (hrs and mins) | Hold | Location | Temp. | Virus Filtered Product Storage Start (Time/Date) | Virus Filtered Product Storage End (Time/Date) | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | R.T. 2-8 °C | ______hrs _____mins | Virus Filtered Product Total R. T. Hold Time: If R.T. Total Hold Time > hrs, contact a supervisor | Virus Filtered Product Total 2-8 °C Hold Time: | Virus Filtered Product Total 2-8 °C + RT Hold Time: If Total 2-8 °C + RT Hold Time > hrs, contact a supervisor

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Product Recirculation Worksheet REUSE

Original BR: VI Att6 · C&D Att7

Record each product recirculation: identify the applicable MPR step & product name, record the recirculation tubing information, then stamp mixing start and end; duration computed (target ≥ 40 min). Assembled from a Record Text + Material Consumption + the Timer block; was mis-flagged 'new' earlier.

Reuses: **DE1 100 stack: SMPL: Record Text Value (Applicable MPR Step / Product Name) + SMPL: Material Consumption (recirculation tubing Part/Batch/Exp/Autoclave) + SMPL: Timer for Start-End Process (field-verified: Start/End Date-Time + Time Difference)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value — recirculation context

Applicable MPR Step *

Product Name *

SMPL: Material Consumption — recirculation tubing SAP Goods Issue · Z_PICONS mvt 261

Tubing Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
				⏱	

SMPL: Timer for Start-End Process — mixing by recirculation (≥ 40 min)

▸ Record

Mixing Start Time *

▸ Record

Mixing End Time *

Mixing Duration (≥ 40 min)

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 4 GROUP(S)):

§22 Attachment 6: Product Recirculation Worksheet VIRUS INACTIVATION

- Obtain a piece of tubing for recirculation of the product. Record the applicable product name and tubing information. Start mixing the product by recirculation. Record the start time. Mix the product by recirculation per SOP-0107097 for at least 40 minutes and record the mixing information.

  Section 7 | Section 8 | Section 9 | Section 15 | Product Name | Tubing Part No. | Tubing Batch No. | Tubing Exp. Date | Tubing Autoclave cycle No. | Tubing Autoclave Exp. Date | Mixing Start Time | Mixing End Time | Mixing Duration * (≥ 40 minutes)

- * The duration of mixing is the time that elapses between the mixing end time and the mixing start time in minutes.

§27 Attachment 7: Product Recirculation Work Sheet CONC / DIAFILTRATION

- Obtain a piece of tubing for recirculation of the product. Record the applicable product name and tubing information. Start mixing the product by recirculation. Record the start time. Mix the product by recirculation per SOP-0107097 for at least 40 minutes and record the mixing information.

  Applicable MPR Step | Product Name | Tubing Part No. | Tubing Batch No. | Tubing Exp. Date | Tubing Autoclave Cycle No. | Tubing Autoclave Exp. Date | Mixing Start Time | Mixing End Time | Mixing Duration * (≥ 40 minutes)

- * The duration of mixing is the time that elapses between the mixing end time and the mixing start time in minutes.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Transfer Tubing / Hosing Information REUSE

Original BR: VI Att5 · C&D Att1

Record each transfer tubing / flex hose: Part No., Batch No. (Exp. auto), autoclave cycle / expiry, hose serial, cleaning batch / expiry. Reused as-is.

Reuses: **DE1 100: SMPL: Material Consumption / SMPL: Additional Assembly (Part/Batch/Exp/Serial/Autoclave)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Material Consumption — transfer tubing (one row per line) SAP Goods Issue · Z_PICONS mvt 261

#	Line / Use	Tubing Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
1	Feed Line					⏱	

+ Add Row

SMPL: Additional Assembly — flex hose (one row per line) SAP Goods Issue · Z_PICONS mvt 261

#	Line / Use	Hose Part No.	Hose Serial No.	Hose Cleaning Cycle No.	Hose Cleaning Exp.	Performed By *
1	Feed Line				⏱	

+ Add Row

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 5 GROUP(S)):

§21 Attachment 5: Transfer Tubing/Hosing Information VIRUS INACTIVATION

- (records)

Tubing/Hose 1 | Tubing/Hose 2, if applicable | Tubing/Hose 3, if applicable | Applicable MPR Step | Tubing Part No, if applicable | Tubing Batch No, if applicable | Tubing Exp. Date, if applicable | Tubing Autoclave Cycle No., if applicable | Tubing Autoclave Exp. Date, if applicable | Hose Serial No., if applicable | Hose Cleaning Batch ID, if applicable | Hose Cleaning Exp. Date, if applicable | Recorded By / Date

§21 Attachment 1: Process Tubing/Hose Information CONC / DIAFILTRATION

- (records)

Hose information for the Retentate Vessel, if applicable | Hose Serial Number | Cleaning Exp. Date | Feed Line | Additional Feed Line, as needed | Retentate Line | Additional Retentate Line, as needed | Permeate Line | Additional Permeate Line, as Needed

- Charge/Diafiltration Line

Additional Charge/Diafiltration Line, if needed | Additional Charge/Diafiltration Line, if needed | Recovery Line | Additional Recovery Line, if needed | Witnessed/Date

- Attachment 1: Process Tubing/Hose Information (continued)

Tubing for the Retentate Vessel, if applicable | Tubing Use | Part Number | Batch Number | Expiration Date | Feed Line | Additional Feed Line, as needed | Retentate Line | Additional Retentate Line, as needed | Permeate Line | Additional Permeate Line, as Needed

- Charge/Diafiltration Line

Additional Charge/Diafiltration Line, if needed | Peristaltic Pump Tubing | Additional Peristaltic Pump Tubing, if needed | Witnessed/Date

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Product Storage & Cross-Record NEW

Original BR: VI 15.8 · C&D §9/§18

Instructions

Record the initial storage condition (RT / 2-8 °C) and stamp the storage start date/time, and cross-record the storage linkage into the Hold-Time attachment / next MPR. Built new: SMPL: Solution Final Storage carries a Storage Location but not the storage-condition + start-date/time this record needs.

Reuses: **Variant of DE1 100: SMPL: Solution Final Storage (adds storage condition RT/2-8 °C + start date/time + cross-record; block has Storage Location only) — no known client variant**

Storage Condition (RT / 2-8 °C) * 2-8 °C

▸ Record

Date *

Time *

Cross-Record — Step / Value written to next MPR *

Performed By *

Witnessed By *

SMPL: Instruction & Sign-off REUSE

Original BR: VI §8/16/20 · C&D §11/20

Reusable instruction-only shell (line/valve connections, dip-tube confirmation, N/A gates, product transfer, BPR review). Text authored per use in IV_INSTR; operator reads and signs. Reused as-is.

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (DE2 903: Long Text Instructions)**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 8 GROUP(S)):

§16 Batch Production Record Review VIRUS INACTIVATION

- (records)
  Production Initial/Date
- Ensure any additional supporting documentation is included (e.g. additional Pi Trends, Memos, Logbook pages, Non-Routine sample requests, etc.).
- Verify that the product net weight after sampling (from Step 15.7) is correctly inputted into SAP.
- Review the Batch Production Record for completeness and submit for review.

§20 Batch Production Record Review CONC / DIAFILTRATION

- (records)
  Production Initial/Date
- If tanks were used, ensure that the SIP, CIP, and Pressure Test Reports for the product tank are attached to the MPR.
- Verify that the PFI product net weight (from Step 18.8) is correctly inputted into SAP.
- Review the Batch Production Record for completeness and submit for review.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Yield Calculations REUSE

Original BR: VI Att4 · C&D Att8

% Yield = Final Product Mass ÷ Load Mass × 100 (masses carry from the weigh / concentration steps). Reused as-is.

Reuses: **DE1 100: SMPL: Yield Calculations**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 4 GROUP(S)):

§20 Attachment 4: Yield Calculations VIRUS INACTIVATION

- (records)

Load Mass Pre-Sampling | Load Volume Pre-Sampling | Load Concentration | Load Mass Pre-Sampling | ___________L_ Step 7.5 | ________g/L__ Step 7.5 | Load Mass Post-Sampling | Load Volume Post-Sampling | Load Concentration | Load Mass Post-Sampling | Load Volume Post-Sampling | Load Concentration | ___________L_ Step 7.12 or Step 8.10 | ________g/L__ Step 7.5 | * Use Step 8.10 volume if Affinity product was transferred to a new VI Treatment Vessel | Final Product Mass Pre-Sampling | Final Product Volume Pre-Sampling | Final Product Concentration | Final Product Mass Pre-Sampling | Final Product Volume Pre-Sampling | Final Product Concentration | ___________L_ Step 14.15 | Final Product Mass Post-Sampling | Final Product Volume Post-Sampling | Product Concentration | Final Product Mass Post-Sampling | Final Product Volume Post-Sampling | Product Concentration | ___________L_ Step 15.7

- Attachment 4: Yield Calculations (continued)

Final Product Mass Pre-Sampling | Load Mass Post-Sampling | Yield | % Yield | Yield | % Yield | Final Product Mass Post-Sampling | Load Mass Pre-Sampling | Yield | % Yield | Yield | % Yield

§28 Attachment 8: Yield Calculations CONC / DIAFILTRATION

- Final Product Mass, Pre-Sampling

PFI Volume before sampling | Final Product Concentration | Final Product Mass, Pre-Sampling (a) | ___________L_ (Step 17.16 or Step 16.33 if no dilution) | ________g/L (8012475-20 or 8012475-30 *) | Volume of the PFI after sampling and after transfer | Final Product Concentration | Final Product Mass, Post-Sampling (b) | ___________L_ (Step 18.9) | ________g/L (8012475-20 or 8012475-30 *)

- * Use (-30) if product was diluted.

Final Product Mass, Pre-Sampling (a) | Total Grams of Virus Filtration Product after sampling (Step 9.10) | Yield | % Step Yield | Final Product Mass, Post-Sampling (b) | Total grams of Virus Filtration Product before sampling (Step 7.8) | Yield | % MPR Yield

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Comments / Deviations Section REUSE

Original BR: VI Att7 · C&D Att9

Free-text log for comments and documented deviations (step no. + initials/date). Reused as-is.

Reuses: **DE1 100: SXS: Phase Comments (no clean 'SMPL: Comments' — closest confirmed component)**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 10 GROUP(S)):

§23 Attachment 7: Comments Section VIRUS INACTIVATION

- (records)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 7: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 7: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 7: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date

§29 Attachment 9: Comments Section CONC / DIAFILTRATION

- (records)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 9: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 9: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 9: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 9: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date
- Attachment 9: Comments Section (continued)
  Step No. | Comments Please initial and date each entry. | Initials/ Date

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Non-Routine Sampling Record REUSE

Original BR: VI Att2 · C&D Att4

Ad-hoc samples requested by MEMO: process section, contents, container, quantity, designation, DLIMS. Reused as-is.

Reuses: **DE1 100: SMPL: Non-Routine Sampling Record**

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 4 GROUP(S)):

§18 Attachment 2: Non-Routine Sampling Record VIRUS INACTIVATION

- If non-routine samples are taken, re-adjust the weight of the product sample.
- Log into DLIMS per SOP-0080295 as required by the memo. Record DLIMS Project number and Sample number if submitting via DLIMS.
  Pre-Load Performed By/Date | Process Section No. | Contents | DLIMS Form | DLIMS Project Number | Container Type | Quantity | DLIMS Tests | Sample Number | Designation | Receiving Group | Storage Temp (°C) | Sample Performed By/Date | Sample Witnessed By/Date

§24 Attachment 4: Non-Routine Sampling Record CONC / DIAFILTRATION

- If non-routine samples are taken, re-adjust weight of the product sampled.
- Log into DLIMS per SOP-0080295 as required by the memo. Record DLIMS Project number and sample number if submitted per DLIMS.
  Pre-Load Performed by/Date | Process Section No. | Contents | DLIMS Form | DLIMS Project Number | Container Type | Sample Volume | DLIMS Test Name | Sample Number | Storage Temp. (oC) | Location

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

Virus Inactivation — PN 8012441 (Low pH, 2000 L)

SMPL: VI Treatment Vessel Setup NEW

Original BR: §8

Instructions

Set up the VI Treatment Vessel. The **selection dropdowns below drive conditional activation** — each gate activates or deactivates the section beneath it, so they are all built into this one XStep. First decide whether the incoming Affinity product vessel is reused as the VI Treatment Vessel (if so, all of Section 8 is N/A). Otherwise obtain & label the vessel (AZD0543 / "VI Treatment Vessel" / Batch / Part / Initials / Date), complete the Bag *or* Tank section, record the vent + product filters, and capture the tare weight with the filters-attached confirmation. **Filters post via Goods Issue (Z_PICONS 261).**

Reuses: **NEW — bespoke single XStep. The Affinity-vessel / Bag / Tank / vent-filter / filters-attached dropdowns DRIVE conditional activation of the sections they gate, and in PI-PCS the activating dropdown must live in the SAME XStep as the sections it activates/deactivates (a dropdown XStep cannot reach down and gate a separate XStep below it). Fields field-verified against SMPL: Select Vessel Type & Tare Weight + SMPL: Material Consumption, but built as one XStep for the branching.**

Use Affinity Product Vessel as VI Treatment Vessel? * No — *activates / deactivates the section below*

Yes → all of Section 8 is N/A and every section below deactivates (continue at Section 9/10). No → obtain & label the vessel and complete the applicable sections below.

Bag Required? * No — *activates / deactivates the section below*

Bag Information (activated when Bag Required = Yes) SAP Goods Issue · Z_PICONS mvt 261

Bag Part No.	Batch No.	Exp. Date	Serial No.	Performed By *

Mixer Drive attached? * Yes / No

Top Base attached? * Yes / No

Magnetic Clamp attached? * Yes / No

Tank Required * No — *activates / deactivates the section below*

Tank Information (activated when Tank Required = Yes)

Tank ID	CIP Batch ID	CIP Exp. Date/Time	Pressure Test ID		Pressure Test Date	Pressure Test Time	SIP Batch ID
				▸ Get Pressure Date/Time			

Vent Filter Required * No — *activates / deactivates the section below*

Vent Filter Information (see Section 4; bold PNs require autoclaving) SAP Goods Issue · Z_PICONS mvt 261

Filter Position	Part No.	Batch No.	Exp. Date	Serial No.	Autoclave Cycle No.	Autoclave Exp.	Performed By *
Vent Filter (if applicable)							

Product Filter Information (see Section 4; bold PNs require autoclaving) SAP Goods Issue · Z_PICONS mvt 261

#	Filter Position	Part No.	Batch No.	Exp. Date	Serial No.	Autoclave Cycle No.	Autoclave Exp.	Performed By *
1	Product Filter 1							

+ Add Row

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Calibration Due Date

Tare Weight of VI Treatment Vessel (kg = L) *

Product Filter attached when tare obtained? * Yes / No

Vent Filter attached when tare obtained? * Yes / No

No filters — vessel is a bag? * Yes / No

Recorded By / Date*

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 13 GROUP(S)):

§8 Transfer Vessel Information and Set up VIRUS INACTIVATION

- NOTE: If the Affinity product vessel will be used as the VI Treatment Vessel, N/A all of Section 8, then continue to Section 9 or Section 10, depending on the Action selected in Step 7.14.
- Obtain the product vessel and label the vessel with “AZD0543”, “VI Treatment Vessel”, batch number, part number, initials, and date.
- For Step 8.4: VI Treatment Vessel Information: Obtain the tare weight of the Treatment Vessel and record the the information in Step 8.4.

  VI Treatment Vessel | Bag Information (if applicable) | Bag Information | Mark the applicable attachments connected to the mixer bag during the vessel tare weight | Attached: | Mark the applicable attachments connected to the mixer bag during the vessel tare weight | Mixer Drive | Top Base | Magnetic Clamp | Tank Information (if applicable) | Tank ID | CIP Batch ID | CIP Exp. Date / Time | Date: | Time: | Pressure Test ID | Pressure Test Date | SIP Batch ID | Filter Information (See Section 4 for acceptable filters and autoclaving requirements.) | Vent Filter (if applicable) | Product Filter 1 | Product Filter 2 | Product Filter 3 | Tare Weight Information | * (NOTE: If collection vessel is a bag, tare weight will be obtained prior to attachment of the product filter.) | Scale/ Balance ID | Calibration Due Date | Tare Weight of the VI Treatment Vessel * | Mark filters attached when tare weight is obtained. | Product filter | Vent filter | No filters, vessel is a bag | Recorded by / Date
- Obtain transfer tubing or a clean flex hose to transfer product from the Affinity Product Vessel to the VI Treatment Vessel. Record the tubing or hose information in Attachment 5: Transfer Tubing/Hosing Information.
- Connect the transfer tubing/hose to the outlet of the Affinity Product Vessel and to the inlet of the VI Treatment Vessel. Thread the transfer tubing through a peristaltic pump, if applicable. Begin transferring the Affinity product to the VI Treatment Vessel.
- After all of the Affinity product has been transferred to the VI Treatment Vessel, disconnect the transfer line from the VI Treatment Vessel.
- Mix the product for at least 15 minutes. Follow SOP-0107102 for the MagMix, if applicable. Follow SOP-0107095 for fixed tanks, if applicable. NOTE: If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.

  Mixer ID/Tank ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥15 minutes)
- Aliquot a 10 mL product sample from the treatment vessel into a 50 mL conical and measure the sample temperature. Record the results below.

  Temperature
- Obtain the gross and net weight of the Affinity product after sampling.

  Scale/Balance ID | Calibration Due Date | Gross Weight/Volume of the Affinity Product after sampling | Net Weight/Volume of the Affinity Product after sampling * | * Net Weight/Volume of the Affinity Product after sampling = Gross Weight/Volume of the Affinity Product after sampling (above) – Tare weight of the Product Vessel (Step 8.4) | Starting Product Temperature Check Proceed per the table below. | Check One | Temperature of the sample at Step 8.9 is: | Action | Within 18 – 25 oC
- Transcribe the pH, conductivity, and temperature results into Step 10.9 “Initial pH / Conductivity / Temperature.” N/A all of Section 9 and continue to Section 10.

  Less than 18 oC:
- Turn on the treatment vessel’s agitator and document the mixing start time in Step 9.1. * Begin warming the vessel per SOP-0080503/SOP-0080023 (for a bag) or SOP-0107095 (for a tank) to reach 18 – 25oC. ** Continue to Section 9.

  Greater than 25 oC
- Contact a supervisor and document actions taken in Attachment 7: Comments Section.
- * NOTE: The product warming range is 18 – 25 oC. ** NOTE: If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Iterative pH Titration Table (Acid / Base) NEW

Original BR: §10 / 12 / 14

Instructions

Iterative pH adjustment by incremental buffer addition. Record initial pH / conductivity / temperature and start time. Per addition: gross weight before & after (net + running total computed), stamp the pH-sample time and record adjusted pH. Target — acid pH 3.5 (NOR 3.4-3.6), neutralization pH 7.2-7.6. Re-standardize the pH meter (4.0/10.0) once pH ≥ 4.0. Buffer consumption posts via Goods Issue (Z_PICONS 261) on the total net weight. Built new — bespoke titration table; the Get Scale / Get pH-Cond Meter / Get Thermometer headers are SMPL: Equipment Select.

Reuses: **NEW — iterative per-addition titration with running totals; no library block. Header uses SMPL: Equipment Select ×3**

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Description

Scale / Balance Cal Due

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

▸ Get Thermometer

Thermometer ID

Thermometer Description

Thermometer Cal Due

Initial pH *

Initial Conductivity *

Initial Temperature (°C) *

Data Input

Addition #	Gross Wt Before (kg)	Gross Wt After (kg)	Net Added (kg)	Total Net (kg)		pH Sample Time	Adjusted pH	Sample Temp (°C)	Performed By *
1					▸ Record	⏱		⏱	

+ Add Row

▸ Record

Final pH Confirmed Time *

Final pH

Final Conductivity *

Final Temperature (°C) *

▸ Record

Incubation Start Time *

Witnessed By *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (3 SECTION(S), 46 GROUP(S)):

§10 Setup and Acid Addition VIRUS INACTIVATION

- As needed, obtain tubing and/or hosing for the transfer of the acid to the starting material. Record the tubing / hosing information in Attachment 5: Transfer Tubing/Hosing Information.
- If using a tank, confirm that the VI Treatment Vessel has a dip tube on the inlet and that the dip tube outlet is below the surface of the product in the treatment vessel.
- Attach tubing/hosing from the outlet of the acid buffer vessel (A-005168) to the inlet of the VI Treatment Vessel, as applicable. Follow SOP-0081015 for a MagMix bag. If applicable, thread the tubing through a peristaltic pump following SOP-0080839. Prime the tubing/hosing leading to the VI Treatment Vessel with (A-005168) pump per SOP-0080839.
- Record the gross weight of the VI Treatment Vessel after priming and the scale/balance used in Step 10.9.
- Ensure agitation is on.
- Calculate the expected amount of acid to be added based on the addition ratio. * Use Step 8.10 if the product was transferred.

  Net Weight of Affinity after sampling (Step 7.12 or 8.10 *) | Acid Addition Ratio (kg acid/ kg Affinity Product) | Recommended weight of acid
- Determine the initial acid addition weight. After initial addition, more acid may be added until the target pH is reached.

  Recommended weight of acid (Step 10.6) | Initial Addition Ratio | Target Weight for Initial Acid Addtion
- Step 10.9: Acid Treatment Begin transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 10.9.

  Acid Treatment | Initial Measurements | Initial pH | Initial Conductivity | Initial Temperature | Start time of addition (End of storage time) | Initial Measurements | mS/cm | Scale/Balance ID | Calibration due date | Gross weight of Treatment Vessel before addition (kg) | End time of acid addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) a | Total net weight of buffer added (kg) b | Time of pH sampling | Adjusted pH | Sample Temperature (oC)
- Note: Do not exceed the volume calculated in Step 10.7 on the initial addition.

  Time of pH sampling | Adjusted pH | Sample Temperature (oC) | 10.9 Acid Treatment Cont. | Time of pH sample confirmed at 3.5 c, d | Final pH (3.4 – 3.6) c | Final Conductivity | mS/cm | Temperature (18 – 25 oC) e | Start Time for Incubation (pH and temperature in-spec) f
- Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).
- Total net weight of buffer added = Net weight of buffer added (kg) +Total net weight of buffer added (kg) from the previous addition. For the first addition only—the previous net weight is zero.
- The target pH is 3.5. The acid addition may be continued after the pH is within range to be closer to the target pH.
- This time should match the time of the first sample where pH is confirmed within 3.4 – 3.6.
- The NOR temperature range for the duration of incubation is 18 – 25 oC.
- Start Time for Incubation (pH and temperature in spec) should match the time of the first sample when the pH is confirmed to be within 3.4 – 3.6 and temperature 18 – 25 oC.

§12 Acid Addition- Extension VIRUS INACTIVATION

- Choose an option based on the Step 11.2 sample result:

  Check One | Condition | Action | The pH is above 3.6 | Continue to the next step. | The pH is within 3.4 – 3.6 and temperature is within 18 – 25 oC | N/A the remainder of this section and Section 13 Go to Section 14. | The pH is below 3.4 or the temperature is not within 18 – 25 oC
- Contact an area manager and document actions taken in Attachment 7: Comments Section.

  Name of Supervisor Contacted, if applicable
- Ensure agitation is on.
- Step 12.4: Acid Treatment Extension Continue transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 12.4.

  Acid Treatment - Extension | Scale/Balance ID | Calibration Due Date | Start time of addition | Gross weight of Treatment Vessel before addition (kg) | End time of acid addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) a | Total net weight of buffer added (kg) b | Time of pH sampling | Adjusted pH | Sample Temperature (oC) | Time of pH sample confirmed at 3.5 c, d | Final pH (3.4 – 3.6) c | Final Conductivity | ------------------------------ mS/cm | Temperature (18 – 25 oC) e | Start Time for Incubation (pH and Temperature in spec) f
- Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).
- Total net weight of buffer added = Net weight of buffer added (kg) +Total net weight of buffer added (kg) from the previous addition. For the first addition only—use the total amount of buffer added in Step 10.9 as the "Total Net Weight of Buffer added."
- The target pH is 3.5. The acid addition may be continued after the pH is within range to be closer to the target pH.
- This time should match the time of the first sample where pH is confirmed within 3.4 – 3.6.
- The NOR temperature range for the duration of incubation is 18 – 25 oC.
- Start Time for Incubation (pH and Temperature in spec) should match the time of the first sample that pH is confirmed to be within 3.4 – 3.6 and temperature 18 – 25 oC.

§14 Setup and Base Addition VIRUS INACTIVATION

- Disconnect the acid transfer tubing from the VI Treatment vessel. Drain the tubing used for acid addition.
- If using a tank and the glycol loop was activated to adjust the treatment vessel temperature, keep the glycol loop on during neutralization to prevent fluctuations to the vessel weight during base addition. If using a bag and a heating blanket was used to adjust the vessel temperature, remove the blanket before proceeding with base addition. If the product did not need to be temperature adjusted, N/A this step and continue to Step 14.3.
- Attach tubing/hosing from the outlet of the base buffer vessel 1M Tris Base (A0101-D) to the inlet of the VI Treatment Vessel, as applicable. If using a MagMix bag, follow SOP-0081015. If applicable, thread the tubing through a peristaltic pump following SOP-0080839. Prime the tubing/hosing leading to the VI Treatment Vessel with (A0101-D).
- Record the gross weight of the VI Treatment Vessel after priming in Step 14.11.
- Ensure agitation is on.
- Calculate the expected amount of base to be added based on the addition ratio.

Total Net Weight of Acid added (Step 10.9 or 12.4 *) | Base Addition Ratio (kg base / kg acid) | Recommended weight of base

- * If additional acid addition was needed, use Step 12.4. Ensure the acid from the first addition is included in the total net weight of acid added.
- Determine the initial base addition weight. After initial addition, more base may be added to reach the target pH.

Recommended weight of base (Step 14.6) | Initial Addition Ratio | Target Weight of Initial Base Addition

- Steps 14.9 and 14.10 provide instructions for filling out the table in Step 14.11 and performing the base additions. Review the instructions in Steps 14.9 and 14.10 before proceeding.
- For Step 14.11 - End of Incubation Information: Take a final sample of the incubation product and measure the pH and temperature. In Step 14.11: Record the (A) ‘pH at the end time of Incubation’ and (B) “Temperature at the end of Incubation’ Record the (C) ‘End time for Incubation’, which is also the start time of the first base addition. Calculate the (D) ‘Total Duration of Incubation’ NOTE: Contact a supervisor if the Total Duration of Incubation exceeds 240 minutes and document actions taken in Attachment 7: Comments Section.
- For Step Step 14.11 - Neutralization Treatment: After a target incubation time of 75 minutes, within a range of 60 – 240 minutes, begin transferring 1M Tris Base (A0101-D) into the VI Treatment Vessel: Do not stop agitation during treatment. For the base additions, follow SOP-0081015 if the VI Treatment Vessel is a MagMix bag. If the vessel is a tank, follow SOP-0107095. After each addition of A0101-D, mix the product for at least ten minutes prior to taking a sample for pH measurement. When pH rises above 4.0, recalibrate the pH meter to pH 4.0 and pH 10.0 and record the calibration information in Step 14.12. Continue adding A0101-D until the product pool is within pH range 7.2 – 7.6. Record the Final pH, Final conductivity, Temperature and End Time For Incubation in Step 14.11.

Neutralization Information | End of Incubation Information | (A) pH at the end time of Incubation (3.4 – 3.6) | (B) Temperature at the end of Incubation (18 – 25 oC) | (C) End time for Incubation (Start Time of base addition) | Start Time of Incubation (Step 10.9 or 12.4) a | (D) Total Duration of Incubation (60-240 minutes) | minutes

Neutralization Treatment | Scale/Balance ID | Calibration Due Date | Gross weight of Treatment Vessel before addition (kg) | End time of base addition | Gross weight of Treatment Vessel after addition (kg) | Net weight of buffer added (kg) b | Total net weight of buffer added (kg) c | Time of pH sampling d | Adjusted pH e

- Note: Do not exceed the volume calculated in Step 14.7 on the initial addition.

Time of pH sampling d | Adjusted pH e | Final pH (7.2 – 7.6) e | Final Conductivity | mS/cm | Temperature (15 – 27 °C)

- Use Step 12.4 if the acid addition was extended.
- Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).
- Total net weight of buffer added = Net weight of buffer added (kg) +Total net weight of buffer added (kg) from the previous addition. For the first addition only—the previous net weight is zero.
- Re-standardization of the benchtop pH meter is needed at Step 14.12 when the pH is ≥ 4.0.
- The target pH after neutralization is 7.4. The NOR pH after neutralization is 7.2 – 7.6.
- Record the standardization of the benchtop pH meter with pH 4.0 and pH 10.0 standards.

pH Meter ID

- Once the product pH has been confirmed to be in neutralization range: Close the outlet of the base vessel and the inlet of the VI Treatment Vessel. Disconnect the base transfer tubing from the VI Treatment Vessel. Drain the tubing used for base addition.
- Obtain and record the gross weight of the VI Treatment Vessel after the neutralization. NOTE: Ensure the VI Treatment Vessel is in the same configuration as when the tare weight was obtained in Step 7.5 (or Step 8.4 if product was transferred to a treatment vessel).

Scale/Balance ID | Calibration Due Date | Gross Weight after Neutralization

- Calculate the net weight of the Neutralized VI product. * Use Step 8.10 if the Affinity Product was transferred to a treatment vessel. ** Use Step 12.4 if additional acid was needed. Ensure that the acid from the first addition is included in the total net weight of acid added.

Net Weight of Affinity Product after sampling (Step 7.12 or 8.10 *) | Total Net Weight of Acid added (Step 10.9 or 12.4 **) | Total Net Weight of Base added (Step 14.11) | Net Weight/Volume of the Neutralized VI Product | kg=L

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Incubation Timing & Incubated Sample (block stack) REUSE

Original BR: §11 / 13

Time the low-pH incubation (start carries from the titration; target 75 min (NOR 60-240)), stamp the mid-incubation sample time, take a mid-incubation temperature check and an incubated pH/cond/temp sample, then make the §13 decision — in-range proceed to neutralization, out-of-range extend incubation and contact the supervisor. Assembled from the Timer block + a Record Text (sample time) + two Equipment Selects + the Solution Summary results block + a Dynamic Dropdown (drives conditional activation) + a Record Text (supervisor). No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Timer for Start-End Process (incubation start/end/duration) + SMPL: Record Text Value (mid-incubation sample time) + SMPL: Equipment Select ×2 (thermometer + pH/Cond meter) + SMPL: Solution Summary - Data Recording (incubated pH/cond/temp) + Dynamic Dropdown (§13 in-range → neutralize / out → extend incubation) + SMPL: Record Text Value (supervisor contacted on extension)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Timer for Start-End Process — low-pH incubation (target 75 min, NOR 60–240)

Incubation Start Time (carries from titration)

▸ Record

Incubation End Time *

Incubation Duration

SMPL: Record Text Value — mid-incubation sample time

▸ Record

Mid-Incubation Sample Time *

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Calibration Due Date

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Calibration Due Date

SMPL: Solution Summary - Data Recording — incubated pH / conductivity / temperature

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High / Low)
1	Incubated pH					

+ Add Row

§13 Decision — pH in range? * In-range *— activates / deactivates the section below*

In-range → proceed to neutralization. Out-of-range → extend incubation (re-time) and contact supervisor.

SMPL: Record Text Value — supervisor contacted (only if incubation extended)

Supervisor Contacted *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (2 SECTION(S), 9 GROUP(S)):

§11 Incubation VIRUS INACTIVATION

- Mid-Incubation Temperature Check for a Bag Vessel If the treatment vessel is a tank, N/A this step. Otherwise, continue below. Incubate the product for a minimum of 38 minutes from Step 10.9 “Start Time for Incubation (pH and temperature in spec),” then sample 10 mL from the treatment vessel. Measure the sample temperature. If the sample is outside the range of 18 – 25 oC, contact a supervisor and document actions taken in Attachment 7: Comments Section.

  Sampling Time | Temperature

- Incubated Product Information Confirm that the product has been incubating for a minimum of 60 minutes. Take a 25 mL sample from the incubated product. Measure the pH, conductivity, and temperature of the sample and record the results below.

  Time of Sampling | pH (3.4 – 3.6) | Conductivity | mS/cm | Temperature (18 – 25 oC)

- Calculate the duration that the product has been incubating.

  Time of Sampling (Step 11.2) | Start Time for Incubation (pH and temperature in spec) (Step 10.9) | Duration the product has been incubating

§13 Incubation - Extension VIRUS INACTIVATION

- Mid-Incubation Temperature Check for a Bag Vessel: If the treatment vessel is a tank, N/A this step. Otherwise, continue below. Incubate the product for a minimum of 38 minutes from Step 12.4 “Start Time for Incubation (pH and Temperature in spec),” then sample 10 mL from the treatment vessel. Measure the sample temperature. If the sample is outside the range of 18 – 25 oC, contact a supervisor and document actions in Attachment 7: Comments Section.

  Sampling Time | Temperature

- Incubated Product Information Confirm that the product has been incubating for a minimum of 60 minutes. Take a 25 mL sample from the incubated product. Measure the pH, conductivity, and temperature of the sample and record the results below.

  Time of Sampling | pH (3.4 – 3.6) | Conductivity | mS/cm | Temperature (18 – 25 oC)

- Calculate the duration that the product has been incubating.

  Time of Sampling (Step 13.2) | Start Time for Incubation (pH and Temperature in spec) (Step 12.4) | Duration that the product has been incubating

- Choose an option based on the Step 13.2 sample result.

  Check One | Condition | Action | The pH is within 3.4 – 3.6 and the temperature is within 18 – 25 oC. | Continue to Section 14

- The pH is not within 3.4 – 3.6 or the temperature is not within 18 – 25 oC.
- Contact an area manager and document actins taken in Attachment 7: Comments Section.

  Name of Supervisor Contacted, if applicable

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Starting / Transfer Temperature Check (block stack) REUSE

Original BR: §7.14 / 9

Measure product temperature and record it. No Dynamic Dropdown is needed: the Record Numeric Value XStep carries Min/Max targets (18–25 °C) and its min/max validation function module enforces the range — an in-range value proceeds to acid addition, an out-of-range value triggers the configured error handling (warm the product if below range / contact supervisor if above). Assembled from Equipment Select + Record Numeric Value. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Equipment Select (thermometer) + SMPL: Record Numeric Value (product temperature, Min/Max 18–25 °C). No decision dropdown — the Record Numeric Value's Min/Max validation function module enforces the range and fires the configured error handling on out-of-range (warm if low / supervisor if high).**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Calibration Due Date

SMPL: Record Numeric Value — product temperature (IV_MIN/IV_MAX = 18–25 °C set on the step; Min/Max validation FM enforces the range + error messaging)

Product Temperature (°C) *

No decision dropdown: the Min/Max validation function module on the Record Numeric Value flags out-of-range and runs the configured error handling (below range → warm the product, Section 9; above range → contact supervisor); an in-range value proceeds to acid addition (Section 10).

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 7 GROUP(S)):

§9 Additional Mixing and Pre-Treatment Product Warming (if applicable) VIRUS INACTIVATION

- Start mixing the product per SOP-0107102 (for a bag) or SOP-0107095 (for a tank) and record the mixing start time. If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.
  Mixer ID/Tank ID | Agitation Setting | Mixing Start Time
- Aliquot a 10 mL product sample from the treatment vessel into a 50 mL conical and measure the sample temperature. Record the results below.
  Temperature | Product Temperature Check Proceed based on the table below. | Check One | Temperature of the sample at Step 9.2 is: | Action | Within 18 – 25 oC
- Measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 “Initial pH / Conductivity / Temperature.” Document the mixing end time in Step 9.4, then continue to Section 10.
  Less than 18 oC
- Continue to warm and agitate the product. * Resample the product regularly until the product is within 18 – 25 oC. When the product is within 18 – 25 oC, measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 “Initial pH / Conductivity / Temperature.” Document the mixing end time in Step 9.4, then continue to Section 10.
  Greater than 25 oC
- Contact a supervisor and document actions taken in Attachment 7: Comments Section.
- * If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.
- Record the mixing end time. Calculate the mixing duration from the time the mixing was started in Step 9.1.
  Mixing End Time | Mixing Duration (≥15 minutes)

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Store VI Product & Hold-Time Link (block stack) REUSE

Original BR: §15.8

Store the neutralized VI product and map values into Attachment 3. This is simply the Product Storage & Cross-Record step + the Hold Time Table, instantiated here — no separate XStep.

Reuses: **Stack of this record's steps: Common - Product Storage & Cross-Record + Common - Hold Time Table, instantiated for §15.8**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

Common - Product Storage & Cross-Record (instantiated for §15.8)

Storage Condition (RT / 2-8 °C) * : 2-8 °C

▸ Record

Date * :

Time * :

Cross-Record — value written to Attachment 3 * :

Common - Hold Time Table (instantiated for §15.8)

Hold #	Location	Storage Temp	Storage Start	Storage End	RT Hold (h)	2-8°C Hold (h)	Performed By *
1		2-8 °C					

+ Add Row

Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)

SMPL: Record CIPDS / Skid / Cassette Information (block stack) REUSE

Original BR: §7.2 / §11 / Att6

Record the UF/DF skid & membrane identity (Skid ID, CIPDS batch, cassette lot/part, support plate, hold-up volume, surface area), confirm the ≤30-day storage window, set up and label the Retentate Vessel — Tank ID, cleaning & sterilization batch/expiry, pressure-test date, vessel volume, vent filter — and capture the retentate tare before and after attachment. Assembled from Record Text Value ×n (collapse to a table) + Material Consumption + Select Vessel Type & Tare Weight (the vessel-setup block) + Equipment Select + Record Scale Weight + Record Numeric Value. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Record Text Value ×n (Skid ID / CIPDS Batch / Cassette Lot / Support Plate / sight-glass / valve labels) + SMPL: Material Consumption (cassette / CIPDS / Retentate Vessel vent filter) + SMPL: Select Vessel Type & Tare Weight (Retentate Vessel: Tank ID + cleaning Batch/Exp + sterilization Batch/Exp + pressure-test date + volume) + SMPL: Equipment Select (scale) + SMPL: Record Scale Weight (retentate tare before & after attachment, Scale ID/Cal Due) + SMPL: Record Numeric Value (hold-up volume / total surface area)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value ×n — skid & membrane identity (collapse to one table)

UF/DF Skid ID *

CIPDS Batch No. *

Cassette Install Date *

No. of Cassettes *

SMPL: Material Consumption — cassette / support plate / Retentate vent filter SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Serial No.	Performed By *
1	Membrane Cassette					

+ Add Row

SMPL: Record Numeric Value — hold-up volume & total surface area

Hold-Up Volume (L) *

Total Membrane Surface Area (m²) *

Cassette storage ≤ 30 days? * Yes — *activates / deactivates the section below*

Confirm the ≤30-day cassette storage window before use.

SMPL: Select Vessel Type & Tare Weight — Retentate Vessel set-up (§11)

Tank ID *

Cleaning Batch / Exp. *

Sterilization Batch / Exp. *

Pressure Test Date *

Vessel Volume (L) *

SMPL: Equipment Select — Scale / Balance

▸ Get Scale / Balance

Scale / Balance ID

Calibration Due Date

SMPL: Record Scale Weight — Retentate tare (before & after attachment)

Tare Weight before attachment (kg) *

Tare Weight after attachment (kg) *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 11 GROUP(S)):

§11 Vessel Set-up CONC / DIAFILTRATION

- Obtain the Retentate Vessel. Label the vessel "Retentate Vessel". Record the vessel's tank ID, clean expiration date and time, sterilization expiration date and time, Cleaning and SIP Batch IDs and date of passed pressure test. Record the volume of the Retentate Vessel.

  Tank ID | Cleaning Exp. Date / Time | Tank Date of Passed Pressure Test | Cleaning Batch ID | Sterilization Exp. Date/ Time | Sterilization Batch ID | Retentate Vessel Volume | Vent Filter Information | Part Number | Batch Number | Expiration Date | Serial Number

- Verify that the vent filter is open on the Retentate Vessel.
- Retentate Vessel Tare Set-Up Label the manual valve on the dip tube of the tank as valve A. Endcap and clamp valve A. Label the tank outlet valve as valve B. End cap the elbow connected to valve B. Label the valve on the tank's retentate tube as valve C. End cap valve C.
- Ensure that the Retentate Tank vent filter is open per Step 11.2, then open Valves A, B, and C all the way (while they are end capped) to relieve pressure between the endcap and the valve outlets. Then, close Valves A, B, and C.
- Record the tare weight of the vessel before it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.

  Scale / Balance ID | Scale/Balance Calibration Due Date | Tare weight of Retentate Vessel before attachment

- Per Attachment 6: Retentate Tank , ensure a sight glass is installed on the following: On the VF Product transfer panel manual valve outlet (valve H) If the VF Product is in a bag, N/A the "Transfer Panel (Valve H)" sight glass fields in this step. Prior to valve A (which is attached to the Retentate Vessel inlet). Prior to valve C (which is attached to the retentate tube on the Retentate Vessel). Record the sight glass information below.

  Sight Glass Information

- Transfer Panel (Valve H), if applicable

  Valve A | Valve C

- Retentate Vessel and UF/DF System Set-Up Set up the Retentate Vessel and DF-1244 system per Attachment 6: Retentate Tank and DF-1244 Set-up. Record the tubing/hose information in Attachment 1: Process Tubing/Hose Information. For tubing, record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. For hoses, record the hose serial number. NOTE: If hoses are used and a serial number is not available, just record that a hose was used in Attachment 1: Process Tubing/Hose Information. Obtain elbows, tees, and manual valves as needed and connect these components to the Retentate Vessel and the UF/DF skid per Attachment 6: Retentate Tank and DF-1244 Set-up. Ensure that manual valves A, B, C, D, E, F, G, H, and I are labeled. Label the Permeate, Retentate, Feed, Recovery, and Charge/Diafiltration (DF) lines.
- Prepare for system priming. Ensure the following valve states:

  Valve States | A, B, C, D, E, F, G, H, I

- Ensure that valves E and H are closed. Initiate the transfer phase on the PCS HMI for the VF Product and all the buffers and the that are coming through the transfer panel. NOTE: Confirm that connections are made in Large Purification before proceeding.

  System Prime: Recovery Line Prime the Recovery Line: Ensure valves A, B, C, D and F are closed. Open bleed valve B-2. Ensure the tubing is directed to waste. Open valves E and G to prime the Recovery Line with the DF Buffer. If necessary, use a pump to prime the Recovery Line. Once no more air is observed exiting bleed valve B-2, close valve E then close bleed valve B-2. Close valve G. NOTE: Buffer charging is performed using diafiltration buffer (PN: 8012555 : 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9)

  System Prime: Charge/Diafiltration (DF) Line Prime the Charge/DF Line: Ensure valves A, B, C, D, and I are closed. Open bleed valve B-1. Ensure the tubing is directed to waste. Open valves E and F from the Diafiltration Buffer to the Charge/DF Line. Use the pump to prime the Charge/DF Line, as necessary. Once no more air is observed exiting bleed valve B-1, close valve E, then close bleed valve B-1. Close valve F. NOTE: Buffer charging is performed using diafiltration buffer (PN: 8012555 : 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9)

- Record the tare weight of the vessel after it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.

  Scale / Balance ID | Scale / Balance Calibration Due Date | Tare weight of Retentate Vessel after attachment | Secure the wheels of the Retentate Vessel. | Tare (zero) the scale holding the Retentate Vessel.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Run Skid Recipe (block stack) REUSE

Original BR: §8/10/13/15/19

Download and run the named skid recipe per phase (Prep-WFI Flush, Equilibration, Conc 1, Diafiltration, Conc 2, Recovery, Clean-WFI Flush, Sanitization): record the Batch ID and recipe setpoints, stamp recipe start and any ≥5-min recirculation (Timer), record retentate weights (Record Scale Weight) and permeate/retentate totalizers, capture the retentate/permeate conductivity & temperature sample results (Solution Summary), log in-process parameters (Operation Monitoring Worksheet) and take the conductivity in-range / re-flush decision (Dynamic Dropdown). WFI-flush phases (§8/§19) first set up WFI — record Bag / Valve / Tank ID and Charge Date/Time and compute the 24-h expiry. Buffer / WFI / NaOH consumed post via Goods Issue (Z_PICONS 261). No bespoke XStep — every field maps to an existing block.

Reuses: **DE1 100 stack: SMPL: Record Text Value (Batch ID + recipe setpoints TMP/crossflow/flush-vol) + SMPL: Record Numeric Value (membrane-area setpoint, permeate/retentate totalizers) + SMPL: Timer for Start-End Process (recipe start + ≥5-min recirculation start/end/duration) + SMPL: Record Scale Weight (retentate weights + Scale ID/Cal Due) + SMPL: Solution Summary - Data Recording (retentate/permeate conductivity & temperature results) + CDF - Operation Monitoring Worksheet (in-process TMP/pressure/flow/UV log) + Dynamic Dropdown (WFI re-flush / conductivity in-range decision) + SMPL: Component Goods Issue (8012555 buffer / WFI / NaOH per phase); §8/§19 add WFI set-up: SMPL: Record Text Value (Bag / WFI Valve-Tank ID + Charge Date/Time) + SMPL: Calc Three Columns (Expiration = Charge + 24 h)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

WFI-flush phase? (§8 / §19) * No — *activates / deactivates the section below*

WFI-flush phases (§8/§19) first set up WFI below; other phases skip to the recipe.

SMPL: Record Text Value — WFI set-up (WFI-flush phases only)

Bag / WFI Valve-Tank ID *

▸ Record

WFI Charge Date & Time *

SMPL: Calc Three Columns — WFI Expiration = Charge + 24 h

Value 1		Value 2		Result
WFI Charge Date/Time	+	24 h	=	WFI Expiration Date/Time

+ Add Row

SMPL: Record Text Value — recipe batch & setpoints

Recipe / Batch ID *

Setpoints (TMP / Crossflow / Flush Vol) *

SMPL: Timer for Start-End Process — recipe start & ≥5-min recirculation

▸ Record

Recipe Start Time *

▸ Record

Recirculation Start *

▸ Record

Recirculation End *

Recirculation Duration (≥ 5 min)

SMPL: Record Numeric Value — membrane-area setpoint & totalizers

Membrane Area Setpoint (m²) *

Permeate Totalizer (L) *

Retentate Totalizer (L) *

SMPL: Record Scale Weight — retentate weight

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Retentate Weight (kg) *

SMPL: Solution Summary - Data Recording — retentate / permeate conductivity & temperature

#	Stream / Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	Permeate Conductivity					

+ Add Row

CDF - Operation Monitoring Worksheet — in-process TMP / pressures / flow / UV log (its own Att-2 card)

Conductivity in range? / WFI re-flush? * In-range *— activates / deactivates the section below*

In-range → proceed. Out-of-range → re-flush and re-sample.

SMPL: Component Goods Issue — buffer / WFI / NaOH consumed this phase SAP Goods Issue · Z_PICONS mvt 261

#	Component	Batch No.	Amount Used (L)	Performed By *
1	Buffer 8012555			

+ Add Row

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (4 SECTION(S), 88 GROUP(S)):

§8 WFI Flush prior to Equilibration CONC / DIAFILTRATION

- Set up per SOP-0107111. WFI collected/obtained directly from the WFI Hold Tank (TK-1458) per SOP-0080303: Verify the SIP of the WFI Hold Tank was completed and the tank status is "Clean." Before connecting to the UF skid, WFI Flush the TK-1458 WFI port on the transfer panel per the following instructions: Ensure an appropriately sized Tee with bleed valve is connected to the manual valve leading from TK-1458 on the transfer panel, then attach a clean hose to the other end of the Tee. Connect the other end of the hose to a process drain port. Ensure any valves on the process drain port are open, if applicable. Open the manual valve leading from the TK-1458 port. Ensure the valve is completely open. Flush TK-1458 through the transfer panel port manual valve to process drain for ≥30 seconds, then close the manual valve on the TK-1458 transfer panel port. N/A the bag information. WFI was/will be collected into a bag: Record the WFI collection vessel information. Verify that the WFI port has been flushed per SOP-0080185. Charge into the vessel and record the charge information.

  Bag, if applicable | WFI Valve ID/Tank ID | WFI Charge Date/Time | Date: | Time: | WFI Expiration Date/Time* | Date: | Time:

- * The WFI Expiration Date/Time is 24 hours from the WFI Charge Date/Time.
- Confirm that the pH meter has been standardized per SOP-0080297. Confirm that the conductivity meter has been standardized per SOP-0080297.

  Meter ID

- Confirm that the thermometer is within its calibration interval. Record the equipment ID of the thermometer used.

  Thermometer ID

- Ensure that the Ultrafiltration (UF) system is set up for WFI flush of the installed cassettes according to SOP-0107111.
- On the UF/DF Skid HMI, enter the total installed cassette surface area in m2 (Step 7.2): Click on the membrane Area Setpoint input box. Enter Total Surface Area of Cassettes in m2 (Step 7.2) and press ENTER. Click ACCEPT on the Membrane Area Change pop-up window.
- Connect the WFI supply inlet to the UF/DF Skid feed line. Ensure that the retentate and permeate lines on the UF/DF Skid are directed to waste.
- Calculate the target volume of solution needed for a 22.0 L/m2 WFI and Equilibration flushes.

  Target Flush Volume | Total Surface Area of Cassettes (Step 7.2) | Target WFI and Equilibration Flush Volume

- Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_PrepWFI_Batch number_Processing Date NOTE: The membranes should be flushed no more than 24 hours before use.

  Batch ID | Start Time

- At the user prompt, input the flush volume set points per the table below.

  Process Step

- Input Title

  Entered Set Point | Permeate Flush | PERMEATE_FLUSH_VOL | ≥ Step 8.7 | Retentate Flush | RET_FLUSH_VOL | ≥ Step 8.7

- Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the pre-use WFI flush. Perform the WFI flush of the UF/DF Skid per SOP-0107111. Sample and document the flush conductivities per Step 8.12.
- At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops.
- At the end of the recipe, take a sample of the retentate and permeate and record the results below.

  Retentate Values | Conductivity (≤ 1 mS/cm) | mS/cm | Temperature (15 – 25°C) | Permeate Values | Conductivity (≤ 1 mS/cm) | mS/cm | Temperature (15 – 25°C)

- If the conductivity is in range, N/A the remainder of Section 8 and go to Section 9. If the conductivity is not in range, go to the next step.
- Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_PrepWFI2_Batch Number_Processing Date. NOTE: The membranes should be flushed no more than 24 hours prior to use.

  Batch ID | Start Time | At the user prompt, input the flush volumes per the table below. | Process Step

- Input Title

  Entered Set Point | Permeate Flush | PERMEATE_FLUSH_VOL | ≥ Step 8.7 | Retentate Flush | RET_FLUSH_VOL | ≥ Step 8.7

- Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the WFI flush of the UF/DF Skid. Perform the WFI flush of the UF/DF Skid per SOP-0107111. Sample and document the flush conductivities per Step 8.18.
- At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops.
- Take a sample of the retentate and permeate and record the results below. If the conductivities are not within the specified range after repeating the recipe, contact an area supervisor and document actions taken in Attachment 9: Comments Section.

  Retentate Values | Conductivity (≤ 1 mS/cm) | mS/cm | Temperature (15 – 25°C) | Permeate Values | Conductivity (≤ 1 mS/cm) | mS/cm | Temperature (15 – 25°C)

§10 Equilibration CONC / DIAFILTRATION

- After the skid WFI flush is completed and prior to proceeding to Equilibration, standardized the skid's UV meter per SOP-0107111.
- Obtain the equilibration buffer, 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 (PN: 8012555). Connect the equilibration buffer to the UF/DF Skid feed line.
- Remove a sample of the equilibration buffer and record the results below.

  Conductivity | mS/cm | Temperature (15 - 25°C)

- Download and run DF1244_EQUIL according to SOP-0107111 to flush the UF membrane(s). Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_Equil_Batch Number_Processing Date.

  Batch ID | Start Time | At the user prompt, input the values per the table below. | Process Step

- Input Title

  Entered Set Point | Permeate Flush | PERMEATE_FLUSH_VOL | ≥ Step 8.7 | Retentate Flush | RET_FLUSH_VOL | ≥ Step 8.7 | Trans-Membrane Pressure (psid) | EQUIL_TMP | Cross Flow (L/min/ft2) c | EQUIL_Crossflow

  The TMP can be set as needed up to NOR of ≤ 35 psid. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure does not exceed NOR of ≤ 60 psig. Cross flow can be adjusted within the NOR range of 0.3 – 0.7 L/min/ft2 as needed to ensure feed pressure does not exceed NOR of ≤ 60 psig. If the cross flow is adjusted, record the new value in Attachment 9: Comments Section. The UF/DF Skid measures the cross flow in L/min/ft2. For conversion reference, L/min/ft2 = L/min/m2 10

- Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the equilibration of the UF/DF Skid. NOTE: Ensure the pH meets the range required per Step 10.8.
- At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops. Record the amount of equilibration buffer 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 flushed using the UF/DF skid's retentate and permeate totalizers. NOTE: The totalizer HMI tags are FQI-301 (retentate) and FQI-201 (permeate).

Retentate Totalizer (FQI-301) | Permeate Totalizer (FQI-201)

- At the end of the recipe, take a sample of the permeate and record the results below.

Permeate Values | Conductivity | mS/cm | Temperature (15 - 25°C)

- If the pH is in range, mark Steps 10.10 through 10.15 as N/A, then proceed to Section 11. If the pH is not in range, go to the next step.
- Download and run DF1244_Equil according to SOP-0107111 to flush the UF membrane(s). Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_Equil2_Batch Number_Processing Date.

Batch ID | Start Time | At the user prompt, input the values per the table below. | Process Step

- Input Title

Entered Set Point | Permeate Flush | PERMEATE_FLUSH_VOL | ≥ Step 8.7 | Retentate Flush | RET_FLUSH_VOL | ≥ Step 8.7 | Trans-Membrane Pressure (psid) | EQUIL_TMP | Cross Flow (L/min/ft2) c | EQUIL_Crossflow

The TMP can be set as needed up to NOR of ≤ 35 psid. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure does not exceed NOR of ≤ 60 psig. Cross flow can be adjusted within the NOR range of 0.3 – 0.7 L/min/ft2 as needed to ensure feed pressure does not exceed NOR of ≤ 60 psig. If the cross flow is adjusted, record the new value in Attachment 9: Comments Section. The UF/DF Skid measures the cross flow in L/min/ft2. For conversion reference, L/min/ft2 = L/min/m2 10

- Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the equilibration of the UF/DF Skid. NOTE: Ensure the pH meets the range required per Step 10.14.
- At the completion of the recipe, the status changes from “Running” to “Complete” and the pump stops. Record the amount of equilibration buffer 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 flushed using the UF/DF skid’s retentate and permeate totalizers. NOTE: The totalizer HMI tags are FQI-301 (retentate) and FQI-201 (permeate).

Retentate Totalizer (FQI-301) | Permeate Totalizer (FQI-201)

- At the end of the recipe, take a sample from the permeate. Record the results below.

Permeate Values | Conductivity | mS/cm | Temperature (15 - 25°C)

- If the pH is in range, go to the next step. If the pH is not in range, contact an area supervisor and document actions in Attachment 9: Comments Section.

Supervisor Contacted

§13 Concentration 1 CONC / DIAFILTRATION

- Ensure the Retentate Vessel is in the same configuration as it was when its Tare Weight after attachment was obtained in Step 11.12.
- To prepare for buffer charge, verify the following valve states on the Retentate Vessel.

A, B, C, D, E, F, G, H, I

- Ensure that skid’s hydraulic valves to the feed (XV101), retentate (XV301 and XV302) and permeate (XV201) lines are closed.
- Open valves A and F. Slowly open valve E from (PN: 8012555) and charge the Retentate Vessel with buffer to achieve the Vessel Charging Volume at 40% (Step 12.2). Once the weight from Step 12.2 has been reached, immediately close valves A, E, and F. • Document the Weight of the Retentate Vessel after buffer charge below.

Scale ID | Cal. Due Date | Weight of Retentate Vessel after buffer charge

- To prepare for recirculation, ensure the following:

A, D, E, F, G, H, I

- On the UF/DF Skid HMI: Open the valves to the feed (XV101), and retentate (XV301 and XV302). Ensure the permeate valve (XV201) is closed. Ensure all drain valves (XV103, XV102, and XV202) are closed. NOTE: Drain valves XV102, XV103, and XV202 should not be opened at any time during processing.
- Recirculate the charging buffer, click on the “Semi-Automatic” mode on the UF/DF skid screen and input the following initial parameters in the order listed:

Feed Pressure (psig) | Feed Valve

- Open

Retentate Valves

- Open

Retentate Valves

- Open

Permeate Valve

- Drain Valves
- Drain Valves
- Drain Valves
- The feed pressure can be adjusted as needed to maintain a target TMP of 20 psid and a maximum AL feed pressure ≤ 60 psig. Record changes to transmembrane pressure and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section.
- Record the start time as the time when the is pump turned on.

Start Time

- Recirculate for at least 5 minutes. At the completion of recirculation, stop the skid pump and record the end time and duration below.

End time for recirculation | Duration of recirculation (≥ 5 minutes) *

- * NOTE: The duration is the time that elapses between the start time for recirculation (Step 13.8) and the end time for recirculation (this step).
- Once buffer recirculation is complete, ensure the following valves on the Retentate Vessel are closed per the table below.

A, B, C, D, E, F, G, H, I

- Ensure that valves E and F from the charge buffer (PN: 8012555) vessel are closed. Ensure that the valve A, H, and I on the Charge/DF Line are closed. Ensure valves to feed (XV101) and retentate (XV301 and XV302) are open. Ensure the permeate (XV201) line is closed.
- Charge the Retentate Vessel with the appropriate amount of Virus Filtered Product according to the charging condition below. Step 9.8: Total VF Product Net Weight after Sampling Step 12.2: Retentate Vessel Charge Volume/Weight at 40% Capacity

Condition | Check one | Action to take | Step 12.2 ≥ Step 9.8

- Charge in all the VF product (Step 9.8)

Step 12.2 ≤ Step 9.8

- Charge to the Retentate Vessel Charge Volume/Weight at 80% Capacity (Step 12.1)
- Prior to charging in the VF product, ensure the VF Product line and the Charge/DF Line are bled of air by ensuring the following valve states:

A, B, C, D, E, F, G | H, I, B1

- Once no more air is observed exiting from B-1, close B-1 and then close valves H and I. Open valve A.
- Open the valve H, then slowly open valve I to charge the appropriate weight of VF product into the Retentate Vessel per Step 13.12, then close valves H and I. Record the weight of the Retentate Vessel after charging and the scale information below. Record the start time of product charge below.

Start Time of Product Charge | Weight of the Retentate Vessel after product charge | Scale/Balance ID | Scale/Balance Calibration Due Date

- Turn on the agitator. Record the agitator speed setting. Adjust the agitation speed as necessary minimizing foaming. Record changes in speed in Attachment 2: Operation Parameters, and rationale for change in Attachment 9: Comments Section.

Agitator Setting

- Prior to product recirculation, ensure the following valve states:

A, D, E, F, G, H, I

- On the skid HMI: Open the valves to the feed (XV101), and Retentate (XV301 and XV302). Ensure the permeate valve (XV201) is closed. Ensure all drain valves (XV103, XV102, and XV202) are closed. NOTE: Drain Valves XV102, XV103, and XV202 should not be opened at any time during processing.
- To recirculate the charged product, click on the “Semi-Automatic” mode on the UF/DF Skid screen and input the following initial parameters in the order listed: The feed pressure can be adjusted as needed to maintain a target TMP of 20 psid and a maximum feed pressure of ≤ 60 psig. Record changes to TMP and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section. Max feed pressure NOR is ≤60 psig.

The start time for recirculation is the time the pump turned on. | Start Time

- Continue to recirculate for at least five (5) minutes. Record the end time and the duration of recirculation. Stop the pump. Ensure valve D is closed.

End time for recirculation | Duration of recirculation (≥ 5 minutes) *

- * NOTE: The duration is the time that elapses between the start time for recirculation (Step 13.19) and the end time for recirculation (this step).
- To prepare for Concentration 1: Open the permeate valve (XV201) on the skid HMI. Ensure the manual valves on the Retentate Vessel are open or closed per the table below.

D, E, F, G, I | A, B, C, H

- Click on the “Add Recipe” button in Recipe Control screen. Find and Select DF1244_CONC from the list. Record the Concentration 1 batch ID and start time from the system overview screen. The batch ID should follow this format: MPR Part Number_Conc_1 _Product Lot Number_Processing Date.

Batch ID | Start Time | At the user prompt, insert the following values. | Process Step

- Input Title

Set Points | Trans-Membrane Pressure (psid) | Initial_Conc_TMP | Cross Flow (L/min/ft2) | Initial_Conc_Crossflow

- The transmembrane pressure can be set up to ≤ 35 psid as needed. If the value is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new value in Attachment 9: Comments Section.
- Push START to initiate the Concentration 1 process. The Start Time is the time the pump turns on.

Start Time

- Concentration 1 Once the skid pump starts, carefully adjust manual valve I on the VF Product Line or adjust the pump speed to control the flow of VF Product into the Retentate Vessel. Maintain the gross charging weight from Step 13.14 until the Virus Filtration Product vessel(s) is empty. Adjust the crossflow and transmembrane pressure (TMP) setpoint(s) from their initial values, as necessary. See Step 13.23, for appropriate ranges. NOTE: If adjustment occurs, record the adjusted values, the time of adjustment, and the step number in Attachment 2: Operation Parameters. Record additional changes to operational setpoints and the time of the change in Attachment 9: Comments Section. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.

Maximum | TMP (psid) | Feed Flow (L/min/ft2) | Feed Pressure (psig) | UV (OD)

- Once the Virus Filtration product vessel is confirmed empty, Close manual valve H on the VF Product outlet. Ensure the following valve states:

D, E, F, G, H, I | A, B, C

- Continue Concentration 1 until the Target product weight in the Retentate Vessel after Concentration 1 (Step 12.5) is achieved. Once the Retentate Vessel reaches the weight in Step 12.5: Stop the pump on the UF/DF skid. Open the retentate valves (XV301 and XV302) completely. Close the permeate valve (XV201).

At the end of Concentration 1, ensure the following: | A, D, E, F, G, H*, I*

- Open
- * NOTE: Valves H and I will remain closed for the rest of the UF/DF process.
- Record the volume of permeate from the permeate totalizer at the end of Concentration 1.

Permeate Volume

- Record the net weight of the product at the end of Concentration 1 from the retentate scale and scale information.

Net Weight of Product after Concentration 1 | Scale ID | Cal. Due Date

§15 Concentration 2 CONC / DIAFILTRATION

- Prior to starting Concentration 2: Reset the permeate totalizer. Ensure the valve states below.

A, D, E, F, G

- To begin Concentration 2, click on the “Semi-Automatic” mode on the UF/DF Skid screen and input the following initial parameters in the order listed. NOTE: Drain valves XV102, XV103, XV202 should not opened at any time during processing. The feed pressure can be adjusted to as needed to a maximum AL feed pressure ≤ 60 psig. Record changes to transmembrane pressure and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section. The TMP can be set to ≤ 35 psid as needed. If the value is adjusted, notify supervisor if TMP is ≥ 35 psid record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. Feed pressure NOR is ≤ 60 psig.

- Push START to initiate the Concentration 2 process. Record the start time as the time the pump turns on.

  Start Time

- Concentration 2: The crossflow and transmembrane pressure (TMP) setpoint(s) may be adjusted from their initial values as necessary as long as they do not exceed the maximum values below. Record the adjusted values, the time of adjustment, and the step number in Attachment 2: Operation Parameters. Record additional changes to operational setpoints and the time of the change in Attachment 9: Comments Section. Record the process data and step number in Attachment 2: Operation Parameters approximately every 20 minutes. Concentration 2 is complete when the retentate weight reaches the Target net weight calculated in Step 12.9. Proceed to the next step for instructions to end Concentration 2. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.

  Maximum | TMP (psid) | Feed Flow (L/min/ft2) | Feed Pressure (psig) | UV (OD)

- When the target weight of the Retentate Vessel after Concentration 2 is reached (Step 12.9): Stop the pump. Ensure the valve states below:

  A, D, F, G Permeate (XV201) | B, C, Retentate (XV301 and XV302)

- Record the end time, retentate weight and the actual volume of permeate from the permeate totalizer after Concentration 2. * NOTE: Ensure the Retentate Weight is within the range calculated in Step 12.9

  Concentration 2 End Time | Retentate Weight * | Permeate Volume

- Recirculation: To recirculate the concentrated product, click on the “Semi-Automatic” mode on the UF/DF Skid screen and input the following initial parameters in the order listed: NOTE: Drain Valves XV102, XV103, XV202 should not opened at any time during processing. The feed pressure can be adjusted as needed to maintain a target of 25 psig and a maximum feed pressure ≤ 60 psig. Record changes to TMP and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section The cross flow can be adjusted, as long as crossflow is ≤ 0.7 L/min/ft2, to obtain the target feed pressure. Do not exceed 60 psig for the feed pressure. If the value is adjusted, record the new value in Attachment 9: Comments Section.

- Recirculate for 5 - 120 minutes (max NOR for recirculation is 120 mins). Record the start and end time of recirculation below. Once recirculation is complete: Stop the pump. Close valves B and C. Close the Feed Valve (XV101) on the HMI.

  Recirculation start time | Recirculation end time | Duration of recirculation (5 - 120 min)

- Ensure the Retentate Vessel is in the same configuration as it was when its tare weight after attachments was obtained in Step 11.12.

- Record the weight of the Retentate Vessel after Recirculation.

  Retentate Weight after Recirculation | Scale ID | Cal. Due Date

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Minimum Membrane Surface Area (block stack) REUSE

Original BR: §7.7

Total Grams Product ÷ Max Loading (g/m²) = Minimum Membrane Area; confirm installed area ≥ minimum. Assembled from a 2-input Calc Three Columns + Record Numeric Value + a Dynamic Dropdown. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns (Total Grams ÷ Max Loading = Min Area) + SMPL: Record Numeric Value (installed area) + Dynamic Dropdown (installed ≥ minimum?)**

▼ *Representative **PI Sheet view** — this is a **block stack** (no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc Three Columns — Minimum Area = Total Grams ÷ Max Loading

Value 1		Value 2		Result
Total Grams Product	÷	Max Loading (g/m²)	=	Minimum Membrane Area (m²)

+ Add Row

SMPL: Record Numeric Value — installed membrane area

Installed Membrane Area (m²) *

Installed area ≥ minimum? * Yes — *activates / deactivates the section below*

Confirm the installed membrane area ≥ calculated minimum before proceeding.

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 8 GROUP(S)):

§7 Initial Product and UF Membrane Information CONC / DIAFILTRATION

- Record the room number where this process will occur.
  - Room Number
- Obtain and record the following information from the AZD0543 CIPDS (FORM-0071789) and Skid Logbook.
  - CIPDS part number | CIPDS lot number | Skid ID | Cassette installation date | No. of cassettes used | Cassette part number | Cassette batch number | Cassette expiration date | Additional cassette batch number, if applicable | Additional cassette expiration date, if applicable | Hold-up volume of the UF/DF Skid | L=kg | Total surface area of cassettes | Support Plate part number | Support Plate batch number | Plate #1 | Plate #2 | Support Plate expiration date | Plate #1 | Plate #2 | Logbook Number | CIPDS SAP Consumption Performed by/Date
- Pre-Processing Activities for the UF/DF Skid Verify the data for the UF/DF Skid to be used. From the logbook, record in Step 7.4 the logbook number, the CIPDS batch number, and the date when the actions were documented. Ensure that the UF/DF Skid's most immediate storage after sanitization has not exceeded 30 days storage. If the storage duration is greater than 30 days, contact a supervisor and document actions taken in Attachment 9: Comments Section.
  - Pre-Processing Activities for the UF/DF Skid | Action | Logbook Number | CIPDS Batch Number | Date | Sanitization | NWP | Storage | DF-1244 Skid Storage Duration (≤ 30 days)
- In the Step 7.6 "Virus Filtration Product Information" table: If a tank was used, N/A Vessels 2 and 3. For bag(s), N/A any unused Vessel fields. Record the net weight of each Virus Filtration Product from (MABR-0027575) MPR or the label of each product transport vessel. Record the tare weight of each Virus Filtration Product Vessel. Record the product concentration of each Virus Filtration Product Vessel. Record the corresponding DLIMS project number and sample number for the Virus Filtration Product(s) vessels.
  - Virus Filtration Product Information | Vessel 1 | Vessel 2, if applicable | Vessel 3, if applicable | Virus Filtration Product Net Weight | Tare Weight | VF Product Concentration | DLIMS Sample Number | DLIMS Project Number | Recorded By/Date
- Calculate the Virus Filtration Product volume in each vessel. NOTE: N/A any unused Vessel fields below.
  - Virus Filtration Product Net Weight (Step 7.6) | Density of VF Product | Virus Filtration (VF) Product Volume | Vessel 1 | 1.014 kg/L | Vessel 1 | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 2, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L | Vessel 3, if applicable | 1.014 kg/L
- Calculate the total grams of Virus Filtration Product in each vessel. NOTE: N/A any unused Vessel fields below.
  - Virus Filtration (VF) Product Volume (Step 7.7) | Virus Filtration Product Concentration (Step 7.6) | Total grams of VF Product per Vessel | Vessel 1 | Vessel 1 | Vessel 1 | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 2, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Vessel 3, if applicable | Total grams of VF Product before sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)
- Calculate the total membrane surface area needed for the UFDF process:
  - Total grams of VF Product before sampling (Step 7.8) | Maximum Load (g of Product / m2) | Minimum membrane surface area
- Verify the total installed surface area of cassettes (Step 7.2) is greater than or equal to the minimum membrane surface area calculated in Step 7.9.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Process Input Calculations (calc-block stack) REUSE

Original BR: §12

Compute the process setpoints from the load (capacity 80%/40%, Conc 1 & 2 target volumes/weights, DF permeate range). A stack of ~9 reusable calc blocks (2-input Calc Three Columns, 3-input Three Variable Calc, and Calc Range Values for the ranges). No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns / Three Variable Calc / Calc Range Values (×~9 setpoint calcs — 80%/40% capacity, Conc 1/2 volumes & weights, DF permeate range)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc stack (~9) — 80%/40% capacity, Conc 1 & 2 volumes/weights, DF permeate range

#	Calculation	Value 1		Value 2		Result (Min / Target / Max)
1	Vessel charge @ 40% capacity	Vessel Volume	×	0.40	=	Charge Volume / Weight

+ Add Row

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 9 GROUP(S)):

§12 Process Input Calculations CONC / DIAFILTRATION

- Calculate the volume at 80% Retentate Vessel Capacity.
  Retentate Vessel Volume (Step 11.1) | Capacity Limit | Retentate Vessel Charge Volume/Weight at 80% Capacity | L = kg
- Calculate the volume at 40% Retentate Vessel Capacity.
  Retentate Vessel Volume (Step 11.1) | Buffer Charge Target | Retentate Vessel Charge Volume/Weight at 40% Capacity | L = kg
- Calculate the product volume in the UF/DF Skid after Concentration 1.
  Total Grams of VF Product after sampling (Step 9.10) | Concentration 1 Range | Product volume in the UF/DF System after Concentration 1 | 110 g/L (max) | L (Min) | 100 g/L (Target) | L (Target) | 90 g/L (min) | L (Max)
- Calculate the product volume in the Retentate Vessel after Concentration 1.
  Product volume in the UF/DF System after Concentration 1 (Step 12.3) | Hold-up volume of the UF/DF Skid (Step 7.2) | Product volume in the Retentate Vessel after Concentration 1 | L (Min) | L (Min) | L (Target) | L (Target) | L (Max) | L (Max)
- Calculate the product weight in the Retentate Vessel after Concentration 1.
  Product volume in the Retentate Vessel after Concentration 1 (Step 12.4) | Density of the Concentration 1 Product | Product weight in the Retentate Vessel after Concentration 1 | L (Min) | kg/L | kg (Min) | L (Target) | kg/L | kg (Target) | L (Max) | kg/L | kg (Max)
- Calculate the permeate volume range for Diafiltration.
  Diafiltration Volume Range (DVs) | Product Volume in the UF/DF System after Concentration 1 (Step 12.3) | Permeate volume range after Diafiltration | L (Min) | L (Target)
- Calculate the product volume range in the UF/DF System after Concentration 2
  Total Grams of VF Product after sampling (Step 9.10) | Concentration 2 Range | Product volume in the UF/DF System after Concentration 2 | 210 g/L (target) | L (Target) | g/L (min) | L (Max)
- Calculate the product volume range in the Retentate Vessel after Concentration 2.
  Product volume in the UF/DF System after Concentration 2 (Step 12.7) | Hold-up volume of the UF/DF Skid (Step 7.2) | Product volume in the Retentate Vessel after Concentration 2 | L (Min) | L (Min) | L (Target) | L (Target) | L (Max) | L (Max)
- Calculate the net weight range of the product in the Retentate Vessel after Concentration 2.
  Product volume in the Retentate Vessel after Concentration 2 (Step 12.8) | Density of the product after Concentration 2 at target | Net Weight of product in the Retentate Vessel after Concentration 2 | L (Min) | kg/L | kg (Min) | L (Target) | kg/L | kg (Target) | L (Max) | kg/L | kg (Max)

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: UF/DF Operation Monitoring Worksheet (Att 2) NEW

Original BR: Att2 (§13-15)

Instructions

Log operating parameters during Concentration 1 / Diafiltration / Concentration 2 using the same conductivity meter as set-up (Step 8.2), including temperature compensation. Per reading record the three transducer pressures (Feed PT-101 / Retentate PT-301 / Permeate PT-201) and computed ΔP, the retentate back-pressure valve opening (BPRV-301) and pump speed (P-101), crossflow flux, TMP (TMP-100), permeate flow (FT-201), permeate flux, total permeate volume (FQI-201), permeate UV (AIT-201), offline permeate conductivity & pH, retentate agitator setting and product weight; the max feed pressure alarm (≤ 60 psig) must not be exceeded. Built new — no library block covers this wide interval log.

Reuses: **NEW — wide interval log (TMP / 3 pressures / permeate flow / conductivity / vessel weight); no library block (Timer-table too narrow, generic parameterized table is a demo). Get Conductivity Meter = SMPL: Equipment Select**

▸ Get Conductivity Meter

Conductivity Meter ID

Conductivity Meter Description

Conductivity Meter Cal Due

Data Input

#		Time	PFeed (psig) PT-101	PRetentate (psig) PT-301	PPermeate (psig) PT-201	ΔP (psid)	Retentate Open (%) BPRV-301	Pump Speed (RPM) P-101	Crossflow (L/min/ft²)	TMP (psid) TMP-100	Permeate Flow (L/min) FT-201	Permeate Flux (LMH)	Total Permeate Vol (L) FQI-201	Permeate UV (OD) AIT-201	Permeate Cond. (mS/cm) *	Permeate pH *	Agitator Setting	Product Wt (kg)	Performed By *
1	▸ Record	⏱																	

+ Add Row

Witnessed By *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 16 GROUP(S)):

§22 Attachment 2: Operation Parameters CONC / DIAFILTRATION

- (records)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.
- Attachment 2: Operation Parameters (continued)

Time | PFeed (psig) @ PT-101 | PRetentate (psig) @ PT-301 | PPermeate (psig) @ PT-201 | ΔP (psid) (= PFeed - PRetentate) | Diafiltration Pump Settings, if applicable | % Retentate Open (%) @ BPRV-301 | Pump Speed (RPM) @ P-101 | Crossflow Flux (L/min/ft2) | TMP (psid) @ TMP-100 | Permeate Flow Rate (L/min) @ FT-201 | Permeate Flux (LMH) | Total Permeate Volume (L) @ FQI-201 | Permeate UV Reading (OD) @ AIT-201 | Permeate Conductivity, offline (mS/cm) * | Permeate pH, offline * | Retentate Vessel Agitator Setting | Product weight in Retentate Vessel (kg)

- * = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Conditional Continued Diafiltration (block stack) REUSE

Original BR: §14

Run the diafiltration recipe and log parameters, record the net weight and compute product volume, then sample the permeate/retentate (pH/cond/temp); if in range proceed, else contact the supervisor, run additional diavolumes and re-sample. Additional 8012555 buffer posts via Goods Issue (Z_PICONS 261). Assembled from the Run Skid Recipe stack + Operation Monitoring + Record Scale Weight + Calc + Equipment Select + Solution Summary + a Dynamic Dropdown + Record Numeric + Record Text. (The decision was published to the PI Sheet Sandbox as a flat form — that stays a representation.) No bespoke XStep.

Reuses: **DE1 100 stack: CDF - Run Skid Recipe (DF1244_Diafiltration batch ID / start / setpoints) + CDF - Operation Monitoring Worksheet (in-process log) + SMPL: Record Scale Weight (net weight after DF + Scale ID/Cal Due) + SMPL: Calc Three Columns (product volume = net wt ÷ density) + SMPL: Equipment Select (pH/Cond meter) + SMPL: Solution Summary - Data Recording (permeate/retentate pH/cond/temp) + Dynamic Dropdown (Continue?) + SMPL: Record Numeric Value (additional diavolumes / permeate totalizer) + SMPL: Record Text Value (supervisor contacted on out-of-range)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

CDF - Run Skid Recipe — DF1244_Diafiltration (batch ID / start / setpoints) — see Run Skid Recipe card

CDF - Operation Monitoring Worksheet — in-process log (its own Att-2 card)

SMPL: Record Scale Weight — net weight after Diafiltration

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Net Weight after DF (kg) *

SMPL: Calc Three Columns — Product Volume = Net Weight ÷ Density

Value 1		Value 2		Result
Net Weight after DF	÷	Density	=	Product Volume (L)

+ Add Row

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Calibration Due Date

SMPL: Solution Summary - Data Recording — permeate / retentate pH / cond / temp

#	Stream / Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	Permeate pH					

+ Add Row

Continue Diafiltration? (pH in range?) * In-range *— activates / deactivates the section below*

In-range → proceed to Concentration 2. Out-of-range → contact supervisor, run additional diavolumes and re-sample.

SMPL: Record Numeric Value — additional diavolumes / permeate totalizer

Additional Diavolumes *

Permeate Totalizer (L) *

SMPL: Record Text Value — supervisor contacted (out-of-range)

Supervisor Contacted *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 27 GROUP(S)):

§14 Diafiltration CONC / DIAFILTRATION

- Diafiltration (DF) Line Priming. Prime the Charge/DF Line with Diafiltration buffer (8012555) by performing the following: First, ensure valves A, D, E, F, and G are closed. Open bleed valve B-1. Ensure the tubing is directed to waste. Open valves E and F to prime the Charge/DF Line. If necessary, use a pump to prime the Charge/DF Line. Once no more air is observed exiting bleed valve B-1, close valve E, then close bleed valve B-1. Close valve F. Ensure that valves B and C are open. NOTE: Ensure no Diafiltration (DF) buffer enters the Retentate Vessel during the prime.
- Click on the “Add Recipe” button in Recipe Control screen. Find and Select DF1244_Diafiltration from the list. Record the Diafiltration batch ID and start time from the system overview screen. The batch ID should follow this format: MPR Part Number_Diafiltration_Product Lot Number_Processing Date.

  Batch ID | Start Time | At the user prompt, enter the following values. | Process Step
- Input Title

  Set Points | Target Permeate Volume after Diafiltration | Diafiltration_Perm_Vol | Step 12.6 | Trans-Membrane Pressure (psid) | Diafiltration_TMP | Cross Flow (L/min/ft2) | Diafiltration_Crossflow
- The TMP can be set to ≤ 35 psid as needed. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new value in Attachment 9: Comments Section.
- Ensure that valves E and F remain closed, then open valve A to prepare for Diafiltration. Push START to initiate the Diafiltration recipe
- Diafiltration: Once the skid pump starts: Open valve E, then slowly open valve F to allow diafiltration buffer to enter the Retentate Vessel. Adjust valve F or the buffer pump speed to control the flow of DF buffer to the Retentate Vessel to maintain the weight from Step 13.30. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. Prior to the end of diafiltration, obtain samples from the retentate and permeate. Record the appropriate readings in Step 14.10. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.

  Maximum | TMP (psid) | Feed Flow (L/min/ft2) | Feed Pressure (psig) | UV (OD)
- At the completion of the Diafiltration recipe once the skid pump stops, close valves E and F.
- Record the permeate volume from the permeate totalizer after Diafiltration and ensure it is within the range calculated in Step 12.6.

  Permeate Volume
- Record the net weight of the product at the end of Diafiltration from the retentate scale and scale information.

  Net Weight of Product after Diafiltration | Scale ID | Cal. Due Date
- Calculate the product volume in the Retentate Vessel after Diafiltration.

  Net Weight of Product after Diafiltration (Step 14.8) | Density of the product after Diafiltration | Product Volume in the Retentate Vessel after Diafiltration | kg/L
- Collect a sample from the permeate and retentate lines for pH, conductivity, and temperature measurement. Record the results below. NOTE: If the Retentate and Permeate pH values are not within range, contact a supervisor prior to proceeding and document actions taken in Attachment 9: Comments Section.

  Retentate Values | Conductivity | mS/cm | Temperature (15 – 25°C) | Permeate Values | Conductivity | mS/cm | Temperature (15 – 25°C)
- Proceed based on the Step 14.10 sample results:

  Check One | The pH of the Step 14.10 retentate samples are: | Actions to take | In-range
- Stop the transfer phase on the PCS HMI for the Diafiltration buffer (PN: 8012555. Close valves A and F. N/A Steps 14.12 through 14.22, then go to Section 15.

  Out-of-range
- Ensure Valves D, E, F, and G are closed. Ensure Valves A, B and C are opened. Go to the next step.
- Contact area supervisor to determine additional volume to be used for coninuted Diafiltration. NOTE: Additional Volume = Target Permeate Volume after Continued Diafiltration NOTE: There is no NOR maximum volume for Diafiltration.

  Additional Volume | Supervisor Contacted
- Click on the “Add Recipe” button in Recipe Control screen. Find and Select DF1244_Diafiltration from the list. Record the Diafiltration batch ID and start time from the system overview screen. The batch ID should follow this format: MPR Part Number_Diafiltration2_Product Lot Number_Processing Date.

  Batch ID | Start Time | At the user prompt, insert the following values. | Process Step
- Input Title

  Set Points | Target Permeate Volume after Diafiltration | Diafiltration_Perm_Vol | Step 14.12 | Trans-Membrane Pressure (psid) | Diafiltration_TMP | Cross Flow (L/min/ft2) | Diafiltration_Crossflow

  The TMP can be set to ≤ 35 psid as needed. If the value is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new the value in Attachment 9: Comments Section
- Ensure that valves E and F remain closed, then open valve A to prepare for Diafiltration. Push START to initiate the Diafiltration recipe.
- Additional Diafiltration: Once the skid pump starts: Open valve E, then slowly open valve F to allow the diafiltration buffer to enter the Retentate Vessel. Adjust valve F or the buffer pump speed to control the flow of DF buffer to the Retentate Vessel to maintain the weight from Step 13.30. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. Prior to the end of Diafiltration, obtain samples from the retentate and permeate. Record the appropriate readings in Step 14.21. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.

  Maximum | TMP (psid) | Feed Flow (L/min/ft2) | Feed Pressure (psig) | UV (OD)
- At the completion of the Diafiltration recipe once the skid pumps stops, close valves E and F.
- Record the permeate volume after continued diafiltration and ensure it is less than or equal to the volume calculated in Step 14.12.

  Permeate Volume
- Record the net weight of the product at the end of diafiltration from the retentate scale and scale information.

  Net Weight of Product after Diafiltration | Scale ID | Cal. Due Date
- Calculate the product volume in the Retentate Vessel after the additional Diafiltration.

  Net Weight of Product after Diafiltration (Step 14.19) | Estimated Density of the Product after Diafiltration | Product Volume in the Retentate Vessel after Diafiltration | kg/L
- Collect a sample from the permeate and retentate lines for pH, conductivity, and temperature measurement. Record the results below.

  Retentate Values | Conductivity | mS/cm | Temperature (15 – 25°C) | Permeate Values | Conductivity | mS/cm | Temperature (15 – 25°C)

- Proceed based on the Step 14.21 sample results:

  Check One | The pH of the Step 14.21 samples are: | Actions to take | In-range

- Stop the transfer phase on the PCS HMI for the Diafiltration buffer. Close valves E and F. Go to Section 15.

  Out-of-range

- Contact an area supervisor and document actions take in Attachment 9: Comments Section.

  Name of supervisor contacted

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Recovery Calculations & Execution (block stack) REUSE

Original BR: §16

Compute the recovery target, choose method (manual/recipe), recover product to the Dilution Vessel, then transfer & mix the recovered product: record the transfer tubing (with hose-cleaning info) and the 'Mixer Drive attached at tare?' flag, stamp transfer end, mix ≥20 min, pull the 8012475-20 sample, weigh the vessel (gross/tare/net) and record concentration + DLIMS, then decide whether dilution is required. Label the PFI mixing bag (SOP-0107056). Recovery buffer posts via Goods Issue (Z_PICONS 261). Assembled from calc blocks + Equipment Select + Material Consumption + Dynamic Dropdown(s) + Run Skid Recipe + Transfer Tubing + Timer + Mixing Time + Sampling + Record Scale Weight + Record Numeric/Text. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns / Three Variable Calc / Calc Range Values (recovery volume/weight range, buffer amount, net/volume after transfer) + SMPL: Equipment Select (mixer + scale) + SMPL: Material Consumption (recovery buffer + Dilution Vessel bag) + Dynamic Dropdown (method manual/recipe + 'Mixer Drive attached at tare?' Yes/No + dilution-required?) + CDF - Run Skid Recipe (recipe-control path) + Common - Transfer Tubing and Hosing Info (transfer line + Hose Cleaning) + SMPL: Timer for Start-End Process (transfer end) + SMPL: Mixing Time (Dilution Vessel mix ≥20 min) + SMPL: Sampling Record + SMPL: Sample Submission Chart (8012475-20) + SMPL: Record Scale Weight (gross/tare/net + Scale ID/Cal Due) + SMPL: Record Numeric Value + SMPL: Record Text Value (supervisor contacted)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc stack — recovery volume/weight range, buffer amount, net/volume after transfer

#	Calculation	Value 1		Value 2		Result (range)
1	Recovery Product Volume Range		×		=	

+ Add Row

Recovery Method? * By recipe control — *activates / deactivates the section below*

Manual → skip the recipe fields. By recipe control → run the recovery recipe below.

CDF - Run Skid Recipe — recovery recipe (recipe-control path; see Run Skid Recipe card)

SMPL: Equipment Select — mixer + scale

▸ Get Mixer

Mixer ID

▸ Get Scale / Balance

Scale ID

Calibration Due Date

SMPL: Material Consumption — recovery buffer + Dilution Vessel bag SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Performed By *
1	Recovery Buffer 8012555				

+ Add Row

Mixer Drive attached at tare? * Yes / No

Common - Transfer Tubing and Hosing Info — transfer line + hose cleaning (see Transfer Tubing card)

SMPL: Timer + SMPL: Mixing Time — transfer end & Dilution Vessel mix (≥ 20 min)

▸ Record

Transfer End Time *

▸ Record

Mixing Start *

▸ Record

Mixing End *

Mixing Duration (≥ 20 min)

SMPL: Sampling Record + Sample Submission Chart — 8012475-20

Sample Designation 8012475-20

▸ Record

Sampling Date & Time *

DLIMS Project / Sample No. *

SMPL: Record Scale Weight — Dilution Vessel gross / tare / net

Gross Weight (kg) *

Tare Weight (kg)

Net Weight (kg)

SMPL: Record Numeric Value — recovered product concentration + DLIMS

Concentration (g/L) *

DLIMS Project ID *

DLIMS Sample ID *

Dilution Required? * No — *activates / deactivates the section below*

Based on the 8012475-20 concentration → dilution required (Section 17) or not.

SMPL: Record Text Value — supervisor contacted (if applicable)

Supervisor Contacted *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 33 GROUP(S)):

§16 Recovery CONC / DIAFILTRATION

- Calculate the Recovery Product volume range.

Total grams of Virus Filtration Product after sampling (Step 9.10) | Recovery Concentration Range | Recovery Product Volume Range | L (Target) | L (Max)

- Calculate the Recovery Product weight range.

Recovery Product Volume Range (Step 16.1) | Density of the Product at Recovery | Recovery Product Net Weight Range | L (Target) | 1.065 kg/L | kg (Target) | L (Max) | 1.065 kg/L | kg (Max)

- Calculate the amount of Recovery buffer to add.

Recovery Product Net Weight range (Step 16.2) | Retentate Weight after Recirculation (Step 15.10) | Weight of Recovery buffer to add | kg (Target) | kg (Target) | kg (Max) | kg (Max)

- Calculate the volume of Recovery buffer to add.

Target Weight of Recovery buffer to add (Step 16.3) | Density of the Recovery buffer | Volume of Recovery buffer to add | 1.008 kg/L

- Prior to Recovery, ensure the valve states listed below:

A, B, F, G, Feed (XV101), Permeate (XV201) | C, D, E, Retentate (XV301 and XV302)

- Choose below whether Recovery will be performed manually or by recipe control:

Check One | Condition | Action to take | Recovery will be run manually | N/A Step 16.7 to Step 16.11. Go to Step 16.12. | Recovery will be run by recipe control | Go to the next step.

Recovery by Recipe Control (Steps 16.7 – 16.11) Click on the “Add Recipe” button on the Recipe Control screen. Select DF1244_RECOVERY_2 from the list. Record the recovery batch ID and start time from the system overview screen. The batch ID should follow this format: MPR part Number_Recovery _Product Batch Number_Processing Date | Batch ID

Start Time | At the user prompt, insert the following values. | Process Step

- Input Title

Set Points | Retentate Volume | Recovery_Ret_Vol | Step 16.4 | Max Feed Pressure | Feed_Pressure_Max | 60 psig a | Target feed pressure is 15 psig. | Push START to initiate the Recovery process.

- Once the pump starts, slowly open manual valve G to allow Recovery buffer to flow into the Retentate Vessel and to maintain a feed flow of less than or equal to 0.7 L/min/ft2. If the feed pressure exceeds 60 psig, stop the process, contact area management, and document actions taken in Attachment 9: Comments Section. Otherwise mark the box below as “NA”

Supervisor Contacted

- At the completion of Recovery when the skid pump stops, immediately close valves C and G to stop the flow of Recovery buffer into the Retentate Vessel. Ensure the remaining manual valves on the UF/DF system are closed per the table below, then N/A Steps 16.12 – 16.13 and proceed to Step 16.14.

A, B, C, D, E, F, G | Manual Recovery (Steps 16.12 – 16.13) Prior to the start of Recovery, verify the valve states below on the UF/DF System: | A, B, F, G, Feed (XV101), Permeate (XV201) | C, D, E, Retentate (XV301 and XV302)

- Open Feed valve XV101 on the skid HMI, then slowly open manual valve G to allow Recovery buffer to flow into the Retentate Vessel and to maintain a feed flow of less than or equal to NOR max of 0.7 L/min/ft2. Monitor the weight of the Retentate Vessel until the target Recovery Product Net Weight (Step 16.2) is achieved, then immediately close valves C and G to stop the flow of Recovery buffer into the Retentate Vessel. Ensure the remaining manual valves on the UF/DF system are closed per the table below.

A, B, C, D, E, F, G

- Obtain the weight of the product after Recovery (with attachments). Record the scale/balance equipment ID and calibration due date.

Weight of product after Recovery (with attachments) | Scale/Balance ID and Cal Due Date | Due Date

- Calculate the amount of Recovery buffer 8012555 used for the Recovery flush.

Retentate Vessel Weight after Recovery (with attachments) (Step 16.14) | Retentate Weight after Recirculation (Step 15.10) | Amount of Recovery buffer 8012555 used for Recovery flush

- Close all the valves on the Retentate Vessel. Disconnect the following parts: The Charge/Diafiltration Line with sight glass, elbow, and tee connected to valve A. NOTE: Keep valve A connected to the tank. The Retentate Line with sight glass and tee connected to valve C. NOTE: Keep valve C connected to the tank. The Recovery and Feed Lines including the reducer and tee connected to the valve B tank outlet. NOTE: This will include disconnecting valve D.
- Place end caps on valves A, B, and C. The Retentate Vessel should be in the configuration per the diagram below.
- Obtain a gross weight of the Retentate Vessel, with no attachments, after Recovery. Ensure that the retentate vessel is the same configuration as in Step 16.17. If necessary, remove the Retentate Vessel from the scale and tare (zero) the scale again before obtaining the gross weight. Record the scale/balance equipment ID and calibration due date.

Gross Weight of Retentate Vessel after Recovery (no attachments) | Scale/Balance ID and Cal Due Date | Due Date

- Obtain a mixing bag for the transfer of the Pre-Formulated Intermediate (PFI) product. Record the part number, batch number, and expiration date for the bag. Ensure the vessel is labeled with “AZD0543”, “Pre-Formulated Intermediate Dilution Vessel”, batch number, 8012475, and initials/date.

Dilution Vessel Info

- Obtain the tare weight for the Dilution Vessel. Record the equipment ID and calibration due date for the scale and the tare weight of the bag. Check if the Mixer Drive is attached when the tare weight is obtained.

Scale/Balance ID | Scale/Balance Calibration Due Date | Tare Weight of Dilution Vessel before attachment | Mixer Drive attached at Tare Weight? (check one) | □ Yes | □ No

- Obtain tubing/hose(s) for the transfer of the product to the Dilution Vessel. Record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. Record the hose cleaning cycle number and expiration date, if applicable.

Tubing Info, if applicable | Hose Info, if applicable | Hose Cleaning Cycle No. | Hose Cleaning Exp. Date | * Autoclave information is not required if tubing is gamma irradiated.

- If the transfer line is tubing, thread the tubing through the peristaltic pump following SOP-0080839.

Turn off the agitator on the Retentate Vessel.

- Connect the Recovery line from the outlet of Retentate Vessel elbow to the inlet of the Dilution Vessel. NOTE: Ensure the Dilution Vessel bottom outlet clamp is closed and clamped as close to the bag as possible.
- Once the Dilution Vessel is connected to the outlet of the Retentate Vessel, open the outlet of the Retentate Vessel at valve B. Pump the entire contents of the Retentate Vessel into the Dilution Vessel and record the end time of transfer below.
- Transfer End Time
- Begin to mix the product by agitation per SOP-0107097. Ensure the agitation is set to mix the product while minimize foaming. Mix for at least 20 minutes. Once a steady agitation rate has been achieved, record the mixer ID, agitation setting, mixing start time, mixing end time and the mixing duration below.

Mixer ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥20 mins)

- Remove product from the Dilution Vessel via a sample port for STAT protein determination (-20) per Attachment 3: Sampling Record. Follow procedures in SOP-0107097.

Sampling Date and Time | Date: | Time:

- Aliquot the sample (sample designation: 8012475-20) per Attachment 3: Sampling Record. If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample with 8012475-20, per SOP-0107056.
- Submit 8012475-20 according to SOP-0080295. Ensure the sampling time and location are updated in DLIMS.
- Remove from the Dilution Vessel the tubing/hosing that was used for transfer. NOTE: Ensure the vessel is in the same configuration as it was when the tare weight before attachment was obtained in Step 16.20.
- Record the gross weight of the Dilution Vessel after transfer. Record the floor scale/balance ID number and calibration due date.

  Gross weight of the Dilution Vessel | Scale/Balance ID | Scale/Balance Calibration Due Date
- Calculate the net weight of the Recovery Product after transfer.

  Gross weight of the Dilution Vessel (Step 16.31) | Tare weight of Dilution Vessel before attachment (Step 16.20) | Net weight of Recovery Product after Transfer
- Calculate the volume of the Recovery Product after Transfer.

  Net weight of Recovery Product after Transfer (Step 16.32) | Density of the product at 190 g/L | Volume of Recovery Product after Transfer | kg/L
- Record the protein concentration and DLIMS Project and Sample ID of the Recovery Product (8012475-20). Based on the 8012475-20 concentration result, indicate below if the product is within the Pre-Formulated Intermediate (PFI) concentration range of 171 – 189 g/L or if it requires dilution.

  Concentration of the Recovery Product | DLIMS Project ID | DLIMS Sample ID | Check One | Concentration | Action | > g/L | Dilution is required: Go to Section 17. | – g/L
- Dilution is not required: N/A Steps 17.1 - 17.16, then go to Step 17.17.

  < g/L
- Contact a supervisor and document actions taken in Attachment 9: Comments Section.

  Supervisor Contacted, if applicable

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Dilution Decision & Calculations (block stack) REUSE

Original BR: §17

Decide whether dilution is required (target ~180 g/L PFI); if so compute the dilution buffer amount, obtain the Intermediate Vessel & transfer tubing, add buffer to target weight (Goods Issue Z_PICONS 261), mix (≥20 min then ≥15 min), pull the -30/-40 samples and record pH/conductivity/temperature, weigh and compute PFI net weight/volume; label the Intermediate Vessel (SOP-0107056). Assembled from Dynamic Dropdown(s) + calc blocks + three Equipment Selects + Material Consumption + Transfer Tubing + Record Scale Weight + Mixing Time ×2 + Sampling + Solution Summary + Record Numeric/Text. No bespoke XStep.

Reuses: **DE1 100 stack: Dynamic Dropdown (dilution required? + dilution method + 'was there dilution?') + SMPL: Calc Three Columns / Calc Range Values (grams, PFI volume/weight range, buffer weight & volume, net/volume after sampling) + SMPL: Equipment Select ×3 (pump / stir plate / scale) + SMPL: Material Consumption (dilution buffer + Intermediate Vessel) + Common - Transfer Tubing and Hosing Info (buffer transfer line) + SMPL: Record Scale Weight (net buffer added, gross after attachments + Scale ID/Cal Due) + SMPL: Mixing Time ×2 (mix ≥20 min then ≥15 min) + SMPL: Sampling Record + SMPL: Sample Submission Chart (8012475-30 / -40) + SMPL: Solution Summary - Data Recording (pH/conductivity/temperature) + SMPL: Record Numeric Value (PFI concentration result) + SMPL: Record Text Value (supervisor contacted)**

▼ *Representative **PI Sheet view** — this is a **block stack** (no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

Dilution Required? * No — *activates / deactivates the section below*

Based on recovered-product concentration. No → N/A the dilution steps. Yes → perform dilution below.

SMPL: Calc stack — grams, PFI volume/weight range, buffer weight & volume

#	Calculation	Value 1		Value 2		Result (range)
1	Weight of Dilution Buffer to Add		×		=	

+ Add Row

Dilution Method? * Connect buffer to vessel — *activates / deactivates the section below*

Pipette / cylinder / carboy → use an Intermediate Vessel. Direct → connect buffer to the Dilution Vessel.

SMPL: Equipment Select ×3 — pump / stir plate / scale

▸ Get Pump

Pump ID

▸ Get Stir Plate

Stir Plate ID

▸ Get Scale / Balance

Scale ID

Calibration Due Date

SMPL: Material Consumption — dilution buffer + Intermediate Vessel SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Performed By *
1	Dilution Buffer 8012555				

+ Add Row

Common - Transfer Tubing and Hosing Info — buffer transfer line (see Transfer Tubing card)

SMPL: Record Scale Weight — buffer added / gross after attachments

Net Buffer Added (kg) *

Gross after attachments (kg) *

Scale ID

Calibration Due Date

SMPL: Mixing Time ×2 — mix ≥ 20 min then ≥ 15 min

Mix 1 Duration (≥ 20 min)

Mix 2 Duration (≥ 15 min)

SMPL: Sampling Record + Sample Submission Chart — 8012475-30 / -40

Sample Designation: 8012475-30 / -40

▸ Record

Sampling Date & Time *

DLIMS Project / Sample No. *

SMPL: Solution Summary - Data Recording — pH / conductivity / temperature

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	pH					

+ Add Row

SMPL: Record Numeric Value — PFI concentration result

PFI Concentration (g/L) *

SMPL: Record Text Value — supervisor contacted (if out of range)

Supervisor Contacted *

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 35 GROUP(S)):

§17 Dilution CONC / DIAFILTRATION

- Calculate the total grams of product in the Retentate Vessel.
  - Volume of Recovery Product after Transfer (Step 16.33) | Concentration of the Recovery Product (Step 16.34) | Total grams of the Recovery Product
- Calculate the AZD0543 Dilution Product Volume Range.
  - Total grams of the Recovery Product (Step 17.1) | PFI Concentration Range | AZD0543 Dilution Product Volume Range | g/L (max) | L (Min) | 180 g/L (target) | L (Target) | g/L (min) | L (Max)
- Calculate the AZD0543 Dilution Product Net Weight Range.
  - AZD0543 Dilution Product Volume Range (Step 17.2) | Density of the Pre-Formulated Intermediate (PFI) | AZD0543 PFI Net Weight Range | L (Min) | 1.062 kg/L | kg (Min) | L (Target) | 1.062 kg/L | kg (Target) | L (Max) | 1.062 kg/L | kg (Max)
- Calculate the weight of Dilution Buffer to Add.
  - AZD0543 PFI Net Weight Range (Step 17.3) | Net weight of the Recovery Product after Transfer (Step 16.32) | Weight of Dilution Buffer to Add | kg (Min) | kg (Min) | kg (Target) | kg (Target) | kg (Max) | kg (Max) | Convert the Weight of Dilution Buffer to Add to volume. | Weight of Dilution Buffer to Add (Step 17.4) | Density of the Dilution Buffer (PN: 8012555) | Volume of Dilution Buffer to Add | kg (Min) | 1.008 kg/L | L (Min) | kg (Target) | 1.008 kg/L | L (Target) | kg (Max) | 1.008 kg/L | L (Max)
- Check the box for the method for how the dilution buffer will be transferred.
  - Check One | Dilution Method | Action
- If using a pipette, grad cylinder, carboy, or bottle
- Proceed to the next step.
- If connecting the dilution buffer directly to the Dilution Vessel
  - N/A Steps 17.7 – 17.9. Go to Step 17.10.
- Obtain a vessel for the transfer of the dilution buffer. Label the vessel as “Intermediate Vessel”, and initial and date. NOTE: N/A any unused spaces.
  - Qty | Carboy | Grad. Cylinder | Bottle | Sterile Pipette | Sterile Pipette | Tubing
- Check the box for the method of addition. If adding using a pump, record the pump ID, if applicable.
  - Pump ID, if applicable | Measurement Method (range reference listed) | Volumetrically Step 17.5 | Weight Step 17.4
- Transfer the dilution buffer (8012555) to the Intermediate vessel.
- If transferring using a pump, connect either the dilution buffer (8012555) or the Intermediate Vessel to the inlet of the PFI Dilution vessel. Thread tubing through a peristaltic pump. Prime the tubing. Otherwise, mark this step as “NA.”
  - Tubing Information | Part Number | Batch Number | Expiration Date | * Autoclave information is not required if tubing is gamma irradiated. | Tare (zero) the scale holding the Dilution Vessel.
- Add dilution buffer to the Dilution Vessel until the target Weight of Dilution Buffer to Add (Step 17.4) is reached. Once target weight is reached, turn off the transfer pump. Record weight after dilution.
  - Net Weight of dilution buffer 8012555 added | Scale/Balance ID | Cal. Due Date
- Disconnect the buffer line from the Dilution Vessel, if applicable. Remove the dilution vessel from the scale and tare (zero) the scale. Obtain the gross weight of the Dilution Vessel after dilution. NOTE: Ensure the Dilution Vessel is in the same configuration as it was when its tare weight was obtained in Step 16.20.
  - Gross weight of Dilution Vessel, with attachments removed | Scale/Balance ID | Calibration Due Date
- Calculate the net weight of PFI.
  - Gross weight of Dilution Vessel, with attachments removed (Step 17.13) | Tare weight of Dilution Vessel before attachment (Step 16.20) | PFI Net Weight
- If the PFI Net Weight is above the max net weight range calculated in Step 17.3,contact a supervisor and document actions taken in Attachment 9: Comments Section. Otherwise, mark this step as “NA”.
  - Supervisor Contacted
- Calculate the volume of PFI after dilution and before sampling.
  - PFI Net Weight (Step 17.14) | Density of the PFI | PFI Volume before sampling | 1.062 kg/L
- Begin to mix the product per SOP-0107097 by agitation. If product is mixed by recirculation, complete Attachment 7: Product Recirculation Work Sheet. Mix the product for at least 20 minutes by agitation. Adjust agitation setting to minimize foaming, as applicable. NOTE: Agitation may already be on.
  - Mixer/Stir Plate ID, if applicable | Agitation Setting | Mixing Start Time | Mixing End Time | Duration of Mixing (≥ 20 minutes)
- Select one of the options below.
  - Check One | Condition | Actions to take
- If there was no dilution
- Mark Steps 17.19 – 17.23 as “NA.” Mark sample (8012475 – 30) in Attachment 3: Sampling Record as “N/A”. Proceed to Step 17.24.
- If there was dilution
- Proceed to the next step
- Remove product from the product collection vessel for (-30) samples per Attachment 3: Sampling Record. Sample per SOP-0107097. NOTE: Ensure the sample line is flushed prior to taking the sample.
  - Sampling Time and Date | Time | Date
- Aliquot the sample (sample designation: 8012475-30) per Attachment 3: Sampling Record If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample per SOP-0107056
- Submit the sample in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.
- Record the results from sample designation, 8012475–30. If the results are not in range, contact supervisor and document actions taken in Attachment 9: Comments Section.
  - PFI concentration (– g/L)

- Once the Pre-Formulated Intermediate has confirmed to be in range, begin to mix the product per SOP-0107097 by agitation. If using a stir plate, record the equipment ID for the stir plate used. NOTE: Agitation may already be on.

  Mixer/Stir Plate ID | Agitation Setting | Mixing Start Time | Mixing End Time | Mixing Duration (≥ 15 minutes)

- Remove product from the product collection vessel for (-40) samples per Attachment 3: Sampling Record. Follow procedures in SOP-0107097 for mixing bags.

  Sampling Time and Date | Time | Date

- Aliquot the sample (sample designation: 8012475-40) per Attachment 3: Sampling Record. If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record.
- Label with 8012475 and additional information per SOP-0107056 Submit the sample in DLIMS per SOP-0080295. NOTE: Ensure the sampling time and location are updated in DLIMS.
- Aliquot product into a 50 ml conical and measure the pH, conductivity, and temperature. NOTE: If the pH value is not in range, contact a supervisor and document actions taken in Attachment 9: Comments Section.

  Conductivity | mS/cm | Temperature (15 – 25 °C)

- Record the gross weight of the Dilution Vessel after sampling. NOTE: Ensure the vessel is in the same configuration as it was when the tare weight was obtained in Step 16.20.

  Dilution Vessel Gross Weight after sampling

- Calculate the net weight of the Pre-Formulated Intermediate (PFI) after sampling.

  Dilution Vessel Gross Weight after sampling (Step 17.30) | Tare weight of Dilution Vessel (Step 16.20) | PFI Net Weight after sampling

- Calculate the volume of the PFI after sampling.

  PFI Net Weight after sampling (Step 17.31) | Density of the PFI | PFI Volume after sampling | 1.062 kg/L

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: PFI Storage Vessel, SHC Filter & Store (block stack) REUSE

Original BR: §18

Obtain & label the PFI storage vessel (bag or autoclaved glass bottle) and SHC filter, connect the transfer tubing, transfer the diluted product through the filter, stamp transfer end, weigh (tare + gross) and compute net weight/volume, then label and store and cross-record hold times into Attachment 5. Assembled from Storage Vessel and Filter Inform + Additional Assembly (bottle) + Material Consumption + Transfer Tubing + Equipment Select + Record Scale Weight + Timer + Calc + Record Numeric + Solution Final Storage + Hold Time Table. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Storage Vessel and Filter Inform (storage vessel + air/vent filter — field-verified) + SMPL: Additional Assembly (glass-bottle autoclave cycle/exp, if bottle) + SMPL: Material Consumption (SHC filter) + Common - Transfer Tubing and Hosing Info (transfer line + Hose Cleaning) + SMPL: Equipment Select (scale) + SMPL: Record Scale Weight (tare + gross after transfer) + SMPL: Timer for Start-End Process (transfer end) + SMPL: Calc Three Columns (net = gross − tare; volume = net ÷ density) + SMPL: Record Numeric Value (net weight) + SMPL: Solution Final Storage + Common - Hold Time Table (Att5 cross-record)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Storage Vessel and Filter Inform — PFI storage vessel + air/vent filter

Storage Vessel (Bag / Bottle) Part / Batch / Exp. *

Air / Vent Filter Part / Batch / Exp. *

Glass bottle used? * No — *activates / deactivates the section below*

If a glass bottle, record its autoclave cycle / expiry below (SMPL: Additional Assembly).

SMPL: Additional Assembly — glass-bottle autoclave (if bottle)

Glass Bottle Autoclave Cycle No. *

Autoclave Exp. Date *

SMPL: Material Consumption — SHC product filter SAP Goods Issue · Z_PICONS mvt 261

Filter	Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
PFI Product Filter (SHC)					⏱	

Common - Transfer Tubing and Hosing Info — transfer line + hose cleaning (see Transfer Tubing card)

SMPL: Equipment Select + Record Scale Weight — tare & gross after transfer

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Tare Weight (kg) *

Gross after transfer (kg) *

SMPL: Timer for Start-End Process — transfer end

▸ Record

Transfer End Time *

SMPL: Calc Three Columns — net = gross − tare; volume = net ÷ density

#	Value 1		Value 2		Result
1	Gross Weight	-	Tare Weight	=	Net Weight (kg)

+ Add Row

SMPL: Solution Final Storage — store PFI

Storage Location *

▸ Record

Date *

Time *

Common - Hold Time Table — cross-record to Attachment 5 (see Hold Time card)

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 13 GROUP(S)):

§18 Filtration and Storage of the Pre-Formulated Intermediate CONC / DIAFILTRATION

- Obtain a mixing bag or bottle for the filtration and storage of the PFI. Record the bag or bottle information below and ensure the vessel is labeled with “AZD0543”, “Pre-Formulated Intermediate”, batch number, part number, and initials/date. NOTE: A mixing bag is preferred to use for storage.

  Bag information (if applicable) | Bag Part No. | Bag Batch No. | Bag Exp. Date | Glass Bottle Information (if applicable) | Glass Bottle Description | Glass Bottle Autoclave Cycle No. | Glass Bottle Autoclave Exp. Date

- Obtain a tare weight of the PFI storage vessel. Record the equipment ID and calibration due date for the scale.

  Scale/Balance ID | Scale/Balance Calibration Due Date | Tare Weight of Storage Vessel

- Obtain an SHC filter per Section 4 and record the filter information below. NOTE: The bolded filter part number(s) in Section 4 require autoclaving.

  Pre-Formulated Intermediate Product Filter Information | Part Number

- Obtain tubing/hose(s) for the transfer of the product to the storage vessel. Record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. Record the hose cleaning cycle number and expiration date, if applicable.

  Tubing Information, if applicable | Hose Information, if applicable | Hose Cleaning Cycle No. | Hose Cleaning Exp. Date | * Autoclave information is not required if tubing is gamma irradiated.

- Attach the outlet of the PFI product filter to the inlet of the PFI storage vessel. Connect the transfer line between the outlet of the Dilution Vessel and the inlet of the PFI product filter. If the transfer line is tubing, thread the tubing through the peristaltic pump following SOP-0080839. Pump the entire contents of the Dilution Vessel into the PFI storage vessel and record the end time of transfer below.

- Transfer End Time

- Drain the hold-up of the PFI product filter and transfer line into the PFI storage vessel. Remove the transfer tubing from the PFI storage vessel and remove the PFI product filter(s).

- Obtain a gross weight of the PFI storage vessel after the transfer. Record the scale/balance equipment ID and calibration due date. NOTE: Ensure the storage vessel is in the same configuration as when the tare weight was obtained in Step 18.2

  Gross Weight of PFI Storage Vessel after transfer | Scale/Balance ID | Scale/Balance Calibration Due Date

- Calculate the net weight of the PFI after sampling and transfer

  Gross Weight of PFI Storage Vessel after transfer (Step 18.7) | Tare Weight of Storage Vessel (Step 18.2) | Net Weight of the PFI after sampling and transfer

- Calculate the volume of the PFI after sampling and transfer.

  Net Weight of the PFI after sampling and transfer (Step 18.8) | Density of the Product after Dilution | Volume of the PFI after sampling and transfer | 1.062 kg/L

- Record the net weight of the PFI (Step 18.8), the tare weight of the PFI storage vessel (Step 18.2), and the gross weight of the PFI storage vessel after transfer (Step 18.7) on the storage vessel label.

- Store the product and document storage information below. For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage. For product stored at 15 – 25°C, the storage start time is the time when final sampling occurred in Step 17.24.

- Record the following values in Attachment 5: Hold Times.

  MPR Name | Name in Attachment | Step No. | Start Time/Date of Product Charge | Virus Filtered Product Storage End (Time/Date) | Step 13.14

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.

SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack) REUSE

Original BR: §19 / §20

Set up WFI (Bag / Valve / Tank ID + Charge Date/Time, 24-h expiry computed), confirm the conductivity meter, enter the membrane surface-area setpoint, run the post-use DF1244_Flush (Clean-WFI), then decide cassette re-use vs discard and run the NaOH sanitization / storage recipe (NaOH via Goods Issue Z_PICONS 261), recording logbook/CIPDS batch, action dates and the buffer batch & amount used (totalizer sum). Assembled from Record Text + Calc + Equipment Select + Record Numeric + Run Skid Recipe + Dynamic Dropdown + Material Consumption + Timer + Text Instructions. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Record Text Value (WFI Bag / Valve-Tank ID + Charge Date/Time) + SMPL: Calc Three Columns (WFI Expiration = Charge + 24 h) + SMPL: Equipment Select (conductivity meter) + SMPL: Record Numeric Value (membrane-area setpoint, permeate/retentate flush setpoints, totalizer amount used) + CDF - Run Skid Recipe (DF1244_Flush batch ID / start) + Dynamic Dropdown (cassette re-use vs discard) + SMPL: Material Consumption (NaOH sanitization / storage buffer) + SMPL: Record Text Value (logbook / CIPDS batch + sanitization/NWP/storage dates + buffer batch) + SMPL: Timer for Start-End Process (start) + SXS: Text Instructions with Sign-off**

▼ *Representative **PI Sheet view** — this is a **block stack** (no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value — WFI set-up

Bag / WFI Valve-Tank ID *

▸ Record

WFI Charge Date & Time *

SMPL: Calc Three Columns — WFI Expiration = Charge + 24 h

Value 1		Value 2		Result
WFI Charge Date/Time	+	24 h	=	WFI Expiration Date/Time

+ Add Row

SMPL: Equipment Select — conductivity meter

▸ Get Conductivity Meter

Meter ID

Calibration Due Date

SMPL: Record Numeric Value — membrane-area & flush setpoints

Membrane Area Setpoint (m²) *

Permeate Flush Setpoint (L) *

Retentate Flush Setpoint (L) *

CDF - Run Skid Recipe — DF1244_Flush (Clean-WFI) batch ID / start (see Run Skid Recipe card)

Cassettes re-used? * Discard — *activates / deactivates the section below*

Discard → NWP not required; clean & store the skid. Re-use → clean, WFI flush, NWP and store.

SMPL: Material Consumption — NaOH sanitization / storage buffer SAP Goods Issue · Z_PICONS mvt 261

#	Buffer	Batch No.	Amount Used (L)	Performed By *
1	0.5 N NaOH (Sanitization)			

+ Add Row

SMPL: Record Text Value — logbook / CIPDS batch + action dates

Logbook No. *

CIPDS Batch No. *

Sanitization / NWP / Storage Dates *

SMPL: Timer for Start-End Process — start

▸ Record

Start Time *

SXS: Text Instructions with Sign-off — BPR review / post-processing (§20)

▼ OLD PAPER BATCH RECORD — VERBATIM REQUIREMENT THIS XSTEP MUST FULFILL (1 SECTION(S), 15 GROUP(S)):

§19 Post-Use WFI Flush CONC / DIAFILTRATION

- Set up per SOP-0107111 using one of the options below. For WFI collected directly from the WFI Hold Tank (TK-1458) Verify the SIP of the WFI Hold Tank was completed and the tank status is “Clean” and that the WFI port has been flushed per SOP-0080303. Mark the bag information as “N/A.” For WFI collected into a bag If the same WFI vessel from Step 8.1 is used: Transcribe from Step 8.1 the WFI Valve/Tank ID, WFI Charge Time/Date and WFI Expiration Date/Time below. N/A the bag information boxes and N/A the SAP consumption box. Record the WFI collection vessel information. Verify that the WFI port has been flushed per SOP-0080185. Charge into the vessel and record the WFI Charge Date/Time and Expiration.

  Bag, if applicable | WFI Valve ID/Tank ID | WFI Charge Date/Time | Date: | Time: | WFI Expiration Date/Time * | Date: | Time:

- * The charged WFI Expiration Date/Time is 24 hours from the WFI Charge Date/Time.
- Confirm that the conductivity meter has been standardized per SOP-0080297.

  Meter ID

- Ensure that the Ultrafiltration (UF) system is set up according to SOP-0107111.
- On the UF/DF Skid HMI, enter the total cassette surface area in m2 (Step 7.2): Click on the membrane Area Setpoint input box. Enter surface area in m2 (Step 7.2) and press ENTER. Click ACCEPT on the Membrane Area Change pop-up window.
- Connect the WFI supply inlet to the UF/DF Skid feed line. Ensure that the retentate and permeate lines to the UF/DF Skid are directed to waste.
- Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_CleanWFI_Batch Number_Processing Date

  Batch ID | Start Time | At the user prompt, input the values per the table below. | Processing Step

- Input Title

  Entered Set Point | Permeate Flush | PERMEATE_FLUSH_VOL | ≥ Step 8.7 | Retentate Flush | RET_FLUSH_VOL | ≥ Step 8.7

- Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the post-use WFI flush.
- At the completion of the recipe, the status changes from “Running” to “Complete” and the pump stops. Continue to Section 20.
- Post-Processing Activities for the UF/DF Skid
- Post-Processing Activities for the UF/DF Skid Check which action will be taken for the cassettes.

  Check One | Decision | Actions to take | The cassettes will not be re-used: Discard cassettes *

- N/A Step 20.2, then continue to Step 20.3 to clean and store the UF/DF skid. * NOTE: If the cassettes are to be discarded, NWP is not required.

  The cassettes will be re-used: Clean, WFI Flush, perform NWP, and store the cassettes | Continue to the next step.

- Verify the data for the UF/DF Skid to be used. From the logbook, record the logbook number, CIPDS batch number and date when the actions were documented below.

  Logbook Number | CIPDS Batch Number | Action | Date | Sanitization | NWP | Storage

- Record the Sanitization and Storage Buffer Batch Number and amount used based on SOP-0107111 * Amount Used = Retentate Totalizer + Permeate Totalizer volume

  Buffer Info | Batch Number | Amount Used (L) * | Sanitization | PN: A0013-D 0.5 N Sodium Hydroxide – DV | Storage | PN: A0014-D 0.1 N Sodium Hydroxide

Instructions shown with the exact fields the operator records (chips), in document order under their real section number.